

Thesis for the Degree of Master of Computer Information System

# **Bias Mitigation in Mental Health Sentiment Analysis using BERT with Fairness Techniques**



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Thesis for the Degree of Master of Computer Information System

**Bias Mitigation in Mental Health Sentiment  
Analysis using BERT with Fairness Techniques**

**Supervised by Prof. Dr. Roshan Chitrakar**

A thesis submitted in partial fulfilment of the requirements for the  
degree of Master of Computer Information System

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**March 202**

## **Dedication**

I dedicate this thesis to my beloved parents, whose endless love, sacrifice, and prayers have shaped every step of my journey; to my dear wife, for her patience, unwavering support, and quiet strength; and to my loving sister, whose encouragement and belief in me have always lifted my spirit.

This work is as much yours as it is mine.

## Declaration

I hereby declare that this study entitled “**Bias Mitigation in Mental Health Sentiment Analysis using BERT with Fairness Techniques**” is based on my original research work. Related works on the topic by other researchers have been duly acknowledged. I owe all the liabilities relating to the accuracy and authenticity of the data and any other information included hereunder.

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## Recommendation

This is to certify that this thesis entitled “**Bias Mitigation in Mental Health Sentiment Analysis using BERT with Fairness Techniques**” prepared and submitted by **Anup Katuwal**, in partial fulfillment of the requirements of the degree of Master of Computer Information System (MCIS) awarded by Pokhara University, has been completed under my supervision. I recommend the same for acceptance by Pokhara University.

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## Certificate

This thesis entitled “**Bias Mitigation in Mental Health Sentiment Analysis using BERT with Fairness Techniques**” prepared and submitted by **Anup Katuwal** has been examined by us and is accepted for the award of the degree of Master of Computer Information System (MCIS) by Pokhara University.

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## ABSTRACT

Mental health care treatment gap around the world requires scalable automated screening, but sentiment analysis models inherit systemic biases of large pre-training corpora on a regular basis. This paper examines bias reduction in three different architectures, namely; BiLSTM with GloVe embeddings (M1), vanilla BERT (M2), and fairness-constrained FairBERT based on adversarial debiasing (M3). The three models were tested on an RMH-Bias-10K dataset that was created on a purpose-built protocol with a five-seed controlled protocol. The RMH-Bias-10K dataset is a stratified dataset of annotated social media posts stratified by gender and dialect subgroups, and the inter-annotator agreement is indicated by Cohen as 88 and 500-post validation subset. Adversarial debiasing of M3 was applied by performing jointly the optimization of a sentiment classifier and a demographic discriminator, where the model was discriminated against whenever its internal representations were keeping demographic information. M3 was more accurate than M1 and M2 by 10.3 and 4.5 percentage points, respectively and improving the mean demographic parity difference and mean equalized odds gap by 35.4 and 22.9 percentage points, respectively, of M2. Although the gains exist there is a 9.5% regression in dialect-based equalized odds and a high false positive rate in female users (0.26 vs. 0.14 in male users) that higher demographic parity does not necessarily solve error-rate differences. These empirical findings support the impossibility theorems of algorithmic fairness and prove that safe and equitable clinical deployment cannot be achieved without subgroup-level analysis, not merely based on the aggregate performance indices.

Keywords: sentiment analysis, BERT, BiLSTM, RMH-bias-10K, DPD, EO-TPR, EO-FPR.

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## List of Abbreviations

| Abbreviation | Fullform                                                |
|--------------|---------------------------------------------------------|
| API          | Application Programming Interface                       |
| AUC          | Area Under the Curve                                    |
| APG          | Accuracy Parity Gap                                     |
| BERT         | Bidirectional Encoder Representations from Transformers |
| BiLSTM       | Bidirectional Long Short-Term Memory                    |
| CDA          | Counterfactual Data Augmentation                        |
| DPD          | Demographic Parity Difference                           |
| EO           | Equalized Odds                                          |
| EOD          | Equalized Odds Difference                               |
| F1           | F1 Score (Harmonic Mean of Precision and Recall)        |
| FPR          | False Positive Rate                                     |
| GPU          | Graphics Processing Unit                                |
| IAA          | Inter-Annotator Agreement                               |
| IEEE         | Institute of Electrical and Electronics Engineers       |
| LMIC         | Low- and Middle-Income Country                          |
| LSTM         | Long Short-Term Memory                                  |
| MCIS         | Master of Computer Information System                   |
| ML           | Machine Learning                                        |

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| NCIT         | Nepal College of Information Technology                                         |
| NHRC         | Nepal Health Research Council                                                   |
| NLP          | Natural Language Processing                                                     |
| OOV          | Out-of-Vocabulary                                                               |
| PRAW         | Python Reddit API Wrapper                                                       |
| RMH          | Reddit Mental Health                                                            |
| RMH-Bias-10K | Reddit Mental Health Bias-Aware 10,000 Post Dataset                             |
| ROC          | Receiver Operating Characteristic                                               |
| TPR          | True Positive Rate                                                              |
| VADER        | Valence Aware Dictionary and sEntiment Reasoner                                 |
| WHO          | World Health Organization                                                       |

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background of the Study

Nearly one billion individuals in the world are impacted by mental health disorders, but, in most areas of the world, the infrastructure established to address the issue merely lacks the capacity to do so [1]. It is acute in low- and medium-income nations where treatment gaps are regularly more than 75% [2]. A case in point is Nepal which is striking. According to the 2020 National Mental Health Survey, 13 percent of Nepali adults have a mental health condition, but only 22.7 per cent of them ever got any type of help, and only 3.7 per cent have ever got some type of hospital-level service [3]. Out of those numbers loom some harsh structural realities: just 0.06 psychiatrists per 100,000 people, divided communities and difficult terrain keeps communities without services apart, and stigma of mental illness makes people reluctant to seek help [3].

Considering these limitations, scholars have begun to consider what can be done with the data that people are already generating. Reddit can be regarded as one of the biggest communities where people share their mental health stories without hesitation and in a shocking fashion that is unlikely to be evoked in a clinical environment [4]. Subreddits such as r/depression, r/SuicideWatch, and r/Anxiety are producing mountains of text every day, and the text they produce contains actual information of distress, coping and recovery. The inquiry that scientists started to make was whether automated tools would be capable of reading that text on a massive level and recognizing individuals who might require assistance [6].

One of the earliest methods that were used on this was Sentiment analysis. Other early tools such as SentiWordNet [7] and VADER [8] used fixed dictionaries where words were given a positive or a negative score. General text dictionaries had been constructed to break down easily on mental health language. The term empty, e.g. is not highly polarized in most lexicons, but in a psychiatric posting it usually denotes intense depression. The sarcasm, the black humor, the evasions of pain all popular in these societies were recorded as neutral or confused the classifiers completely [9]. Recurrent architectures, and BiLSTM networks with

GloVe embeddings [11] in particular, were an obvious improvement since they were not trained on rules, but on labeled data. They have been applied to mental health text with studies reporting meaningful improvements over lexicon baselines [5], [12] although they still apparently had issues with the emotional density of crisis-level writing.

Models based on transformers altered the image on a deeper level. BERT [14] that is pre-trained on large text sets and fine-tuned to particular tasks became superior to everything that preceded it in NLP benchmarks, and mental health researchers hastened to implement it. All of depression detection [15], suicide risk screening [16], and emotional state classification [17] improved in terms of gains when BERT substituted previous models. The trend was regular enough across research works that BERT became the default starting point of this type of work [14], [15], [16].

The challenge is that BERT does not only learn language. It acquires the language of its training data with its inherent demographic skews. It has been reported by a number of studies that BERT-family models generate systematically different outputs when given text connected to dissimilar demographics [20], [21], [22], [23]. In mental health screening, the situation dramatically changes: when a model systematically underestimates distress in posts made by women, or by text that is not written in standard English, then these users are missed. There is no backup. Nepal just makes that point sharper, in that there are so few professionals that an automated system that is biased will not be able to depend on the human review of the downstream to identify its mistakes [24], [25].

Although there is an increasing interest in algorithmic fairness in NLP [26], [27], the majority of it has focused on high-resource Western settings. Such methods as adversarial debiasing [28] and counterfactual data augmentation [29] had been proven to be useful in general NLP tasks, and none of them had been tried on mental health text before this paper. The use of language in mental health is also emotionally saturated, setting, and disproportionate according to the severity of the crisis in a manner that is not reflected in a general NLP standard. It was of true unknown whether bias mitigation approaches can be transferred to this area.

## 1.2 Problem Statement

Sentiment analysis with BERT is effective by standard measures on mental health text, though the measures conceal a fairness issue. These models have demographic biases in their pre-training, resulting in unequal predictions across groups of users- reported examples are gender bias [20], inconsistent dialect variation prediction [26], [27] and false association between demographic cues and sentiment prediction [23]. In a mental health setting, such imbalance is not an abstract statistic issue. It implies that two posts with similar levels of distress will be given different model results when written by different people, which in the real world is equivalent to disproportionate detection of at-risk people and unequal distribution of resources that are already limited. It is in Nepal, and in Nepal such as only 22.7% of individuals with mental health issues get any form of treatment [3], and clinical supervision is limited, where a systematically biased tool has no gap - it merely adds to the disparities already present.

The problem is difficult to solve because of two gaps. Current datasets of mental health sentiment [16], [17], [31] were not constructed in a way that would provide fairness and would not have the annotations of demographics to at least gauge whether a model is acting fairly or not. In the absence of such data, the question cannot be posed. Simultaneously, methods of bias mitigation, such as adversarial debiasing [28] and counterfactual data augmentation [29], have not been experimented with mental health text, where the imbalance between classes is the greatest, and the impact of misclassification is the largest. The thesis fills both gaps: it creates RMH-Bias-10K, 10,000 posts dataset with demographic annotations on four attributes that have protection, and empirically tests whether the disparities can be reduced with FairBERT (M3) without compromising the classification quality.

## 1.3 Research Objectives

The main aim of the proposed study is to create and empirically test fairness-conscious sentiment analysis models to apply to mental health problems with the help of Reddit data. This goal is supported by three particular objectives:

- To create RMH-Bias-10K, 10,000 mental health posts with binary sentiment tags and demographic characteristics were annotated by all four dimensions with the

following categories that are inferred: inferred gender, crisis severity, dialect variation, and subreddit affiliation.

- To compare the results achieved in sentiment classification and fairness by three model structures: BiLSTM and GloVe embeddings (M1), vanilla BERT (M2), and FairBERT using adversarial debiasing and counterfactual augmentation (M3).
- To test the hypothesis of whether adversarial debiasing and counterfactual data augmentation minimize demographic biases on all four of the protected attributes without a significant decrease in classification accuracy.

#### **1.4 Significance of the Study**

The relevance of this work lies at the crossroad of two issues, which have not been examined simultaneously. NLP algorithmic fairness has evolved at a high pace, yet mental health sentiment analysis has largely been left out of it. This paper is the point of contact between those two spheres and generates empirical data where there has been largely theoretical interest.

Theoretical contribution is a fairness assessment performed in the mental health NLP in particular. This is important in that the field possesses characteristics that make it unlike the general NLP scenarios most debiasing has been performed in: extreme imbalance in classes, language that is emotionally charged, vocabulary that the community relates to, and high stakes when it comes to misclassification. These conditions give evidence on the interaction between adversarial debiasing, counterfactual augmentation and focal loss, and the achieved performance and fairness improvements under these conditions undercut the popular belief that we need to compromise predictive accuracy to increase demographic equity.

RMH-Bias-10K methodologically addresses a gap in literature that has not been addressed. As far as I understood, there is no previous data on mental health sentiments that contains multi-dimensional annotations of demographics specifically created to assess fairness. The annotation pipeline that was created in this work, with the inter-annotator agreement metric tested on a trained mental health practitioner, is described sufficiently to be replicated or scaled to other sensitive tasks.

Its practical significance is most immediate to Nepal, but is beyond that country. The whole pipeline of the experiment is implemented in Google Colab using open-source libraries,

which implies that the method is available in resource-limited environments. The observation that M3 can minimize disparities in demographics, but it does not affect the accuracy of the classification is an empirical conclusion that can make any difference in the deployment decision-making in the context of the public health issue, although it is vital to realize that the human oversight should be an inseparable part of any real-world system. The same problem that is tackled in this thesis is experienced in South Asia more generally, as well as in any other setting that has a combination of linguistic diversity, limited clinical capacity, and an increasing uptake of social media.

### **1.5 Scope and Limitations**

The dataset contains English-language posts of 10 mental health communities on Reddit, including r/depression, r/Anxiety, r/mentalhealth, r/SuicideWatch, r/BipolarReddit, r/PTSD, r/OCD, r/lonely, r/selfharm, and r/offmychest, and was gathered in the period between January 2020 and December 2024. These subreddits were chosen due to size, level of activity and conditions covered. Publications written in other languages, content on other platforms, and official clinical documentation are out of the scope of the study. The quality of the findings to be generalized to the platforms with other user demographics and content norms remains an open question.

There are four attributes that are evaluated by fairness, which include inferred gender (male/female), crisis severity (high/low), dialect variation (standard/non-standard English), and subreddit affiliation. Other characteristics that might be of interest age, ethnicity, socioeconomic background, particular diagnoses are not annotated, in part because they are hard to determine reliably using text and in part due to privacy considerations [32]. It is the gender annotation in particular which is binary and does not represent the entire spectrum of gender identities. The inference method is connected to that limitation, and the findings based on gender are to be interpreted with such limitation in mind.

This model comparison will be limited to three architectures: M1 (BiLSTM with GloVe), M2 (vanilla BERT) and M3 (FairBERT) architectures. Other fairness interventions, such as reweighting or post-hoc calibration, like RoBERTa [18] and XLNet [19] are not in the scope. Adversarial debiasing was ranked highest due to the fact that it can be incorporated directly into the training and demonstrated computational efficiency in previous studies [28] and thus

it was an efficient option in Colab-based experimentation. Assessment is based on conventional classification measures, accuracy, precision, recall, F1, and AUC-ROC, but not clinical risk measures based on domains. It is entirely a retrospective study; there was no real-time tracking of the study, no contact with the users, or deployment of any kind.

## **1.6 Organization of the Thesis**

The rest of this thesis is structured in the following way. Chapter 2 summarizes the literature on the sentiment analysis methods, mental health applications, bias reduction in language models, and mitigation of bias. Chapter 3 explains how the RMH-Bias-10K dataset has been constructed, data has been collected, annotated, and the validation of inter-annotator agreement. Chapter 4 introduces the research methodology, including the description of the three model architectures and training set-ups. The empirical findings have been reported in chapter 5; these include classification performance measures and analysis of fairness metric on the basis of demographic attributes. Chapter 6 puts the results into perspective of the research goals, implications of the research on fairness-aware systems deployment, limitations of the study, and future research directions.

## **CHAPTER 2**

### **LITERATURE REVIEW**

#### **2.1 Overview**

This chapter discusses three areas that are used to inspire and contextualize experiments in subsequent chapters. The former is the history of the methods of sentiment analysis - how the area developed to focus on hand-crafted dictionaries, and then deep learning, and transformers, and what was sacrificed in the process. The second one is the study that transposes those approaches to mental health text on social media, which shows what can be done and where the issues lie. The third one is the fact that the large models that are pre-trained have demographic biases and the methods researchers have developed to mitigate them. The focus of the review is on work published in 2018 up to 2025, at which point models based on transformers are the mainstream, even though earlier foundational research is covered where it still applies to what this thesis does.

#### **2.2 Sentiment Analysis Techniques in Natural Language Processing**

Sentiment analysis Sentiment analysis Attempting to identify the subjective attitudes and emotions of a text [41] has experienced a number of different stages throughout its 20-year history. The motivation of each transition was not only by improved performance on benchmarks but by alterations in what the models could actually perform with language.

The first tools were constructed based on the manually made dictionaries. WordNet entries were given polarity scores by SentiWordNet [7], and VADER [8] introduced capitalization, punctuation, negation rules to enable it to differentiate between good and GOOD!!!!. and deal with such phrases as not great. These systems did not need any labeled training data and were readily inspectable and explainable. On serious, well-cut copy they performed fairly well. The issues were brought up soon on more messy material. The dictionary also gives the word an identical score in any scenario, thus empty is said to be neutral even when being used to define a psychological condition. Sarcasm, dark humor, and indirect manifestations of distress, which often have an online mental health group, are either

ostensibly neutral or are mislabeled entirely [9]. Procurement of these tools into a different field entails manual reestablishment of the dictionary and this is not scalable.

Statistical machine learning transformed the method essentially with the learning of labeled instances as opposed to hand-written regulations. The word frequency-based naive Bayes classifiers formed a common baseline following the initial sentiment analysis efforts on movie reviews [41]. n-gram based Support Vector Machines performed well on formal benchmarks and were reliably better than rule-based methods. The assumption behind this, however, was bag-of-words: words are counted, but their sequence is not taken into consideration. This complicated the negation (not happy), intensification, and changes in tones in a single sentence.

Recurrent neural networks handled the sequence problem in a direct manner. The networks of Long Short-Term Memory [10] have internal states held across a sequence, and therefore can pass information about the past tokens to the upcoming ones. Their gating mechanisms, input, forget and output gates regulate what is stored and what is forgotten and also helped address the vanishing gradient issue that had plagued prior instances of recurrent networks to train long sequences. Bidirectional LSTMs [11] process the input twice, in a forward and a backward pass, and therefore every point in the input has forward and backward context. Combining BiLSTMs with pre-trained GloVe word embeddings [11], which are vectors trained on billions of tokens representing semantic similarity, showed significant improvements over bag-of-words baselines on mental health classification tasks [5], [12]. The model can be augmented with attention mechanisms [44], where the positions are weighted based on their importance to the prediction as opposed to all positions. Another state of the art in mental health text classification was BiLSTM with attention, which operated between approximately 2015 and 2018 and is an M1 baseline in the paper.

Transformers [13] did not use the sequential order of recurrent models. A transformer does not calculate attention between successive tokens of a hidden state, but every pair of positions simultaneously using scaled dot-product attention. There are several attention heads that operate concurrently each of which may learn to attend to various kinds of relationships in the input. Deep stacking can be performed normally without training instabilities by residual connections and layer normalization [46]. The computational cost is

significantly lower than recurrent models: transformers can be parallelized over the sequence length, and so training on big data is much faster.

This architecture was used in pre-training of large scale by BERT [14]. BERT was pre-trained on about 3.3 billion words of English Wikipedia and the Book Corpus using masked language modeling, in which random tokens are substituted with a mask symbol and the model is trained to predict them based on context, as well as next-sentence prediction. The representations were then principled by fine-tuning them on a task with small volumes of labeled data. At the time of its release, BERT set state-of-the-art scores on eleven NLP benchmarks and became the default initial point in transfer learning in NLP. BERT-based models outperform BiLSTM models by five to ten macro-F1 points on typical benchmark tasks on sentiment classification in particular [13], with larger differences in linguistically challenging tasks. The pre-training and fine-tuning structure is figured in Figure 2.2.

Subsequent versions found flaws with the first training configuration of BERT. It was demonstrated by RoBERTa [18] that BERT was undertrained significantly, and large gains were attained by eliminating the next-sentence prediction task, increasing batch sizes and training time, using longer sequences, and employing dynamic masking. XLNet [19] came up with an objective that uses permutation to retain bidirectional context without the use of masked tokens. Both models further advanced the performance and have been applicable in the classification of mental health. In this paper, BERT-base is the chosen model of transformers since it is the most studied and reported. Chapter 6 admits that RoBERTa and XLNet are not included in the comparison.

The application of any of these models to the mental health posts in Reddit brings on some challenges that fail to be reflected by the usual NLP benchmarks. The text is frequently disjointed, emotionally charged, and ambivalent - medical jargon and internet terms, explicit and explicit or highly coded expression of distress. Vocabulary that is specific to communities is fast evolving and is mostly not represented in pre-training corpora. A BERT model trained on Wikipedia and news articles will fail to capture much of this unless it is carefully trained and evaluated considering the patterns that are specific to the domain.

## 2.3 Applications of Sentiment Analysis in Mental Health Monitoring

Mental health monitoring through computational analysis of social media text is a theory that has been actively researched since the early 2010s. The rationale is that the patterns of the language used on the user posts bear quantifiable cues concerning the mental state, and, therefore, passive screening on a large scale is at least theoretically achievable. There is a large body of evidence attesting to this point, but the literature also indicates that the method is also associated with actual constraints that are down-played in initial positive discussions.

Most of the initial data was given by Twitter. Coppersmith et al. [12] demonstrated that linguistic characteristics were sufficient to separate self-reported depressed users and controls matched at approximately 70% accuracy. De Choudhury et al. [5] conducted a longitudinal study and discovered changes in language, frequency of postings, and social interaction that can be measured in the months preceding the initial diagnosis of depression in a user. The trends were similar: the increase in the use of the first-person singular pronoun, the negative-emotion words, the reduction in the sentence length, and the reduction of the discussion of social subjects. The research of these studies showed that the signal is there and this attracted extensive research that followed.

Reddit had become the site to do more in-depth work. Compared to the concise messages sent on Twitter, long-form posts offer more linguistic data, and mental health subreddits are structured into particular conditions, which offer researchers fairly interpretable labels depending on the membership of a community. Since 2014, the CLPsych shared task series [16], held regular competitions on Reddit mental health data, such as suicide risk prediction in r/SuicideWatch and depression classification in r/depression. These activities developed standard benchmarks that enabled the comparison of the results of research groups. The task [16] of 2019 involved categorizing users into four levels of suicide risk; optimized systems used BERT encoders with behavioral metadata (posting frequency) and further gains were achieved with ensemble techniques. The four-class task with Macro-F1 was 0.50 to 0.65, which indicates not only the actual ability of the model, but also the actual hardness of the fine-grained risk classification using text only.

There is also an increase in emotion classification research, which is partially motivated by increased datasets. GoEmotions was published by Demszky et al. [17], containing 58, 000

Reddit comments that are categorized in 27 emotion categories. Although not mentally health-specific, it has been applied to pre-train emotion classifiers then adapted to clinical tasks which is an effective transfer learning method when the number of labeled mental health examples is limited.



Fig. 2.1: Evolution of Sentiment Analysis Methods

Some nagging issues pervade this literature. The steadiest one is perhaps the balance of classes: although during a crisis, even in crisis-oriented communities, grave, or sharp posts constitute a minor percentage of the overall number. This is because a model that only learns to predict the majority class can get misleadingly high accuracy but cannot be used to identify the at-risk individuals that it is designed to identify. This is concealed by standard aggregate measures. The way this study addresses the problem is outlined in Section 4.4.

The second big problem is the cultural and linguistic peculiarity. The pre-trained models mirror the language of the training data that is skewed towards formal Standard American or British English. Community-specific vocabulary, non-standard dialect phenomena and culturally-influenced manifestations of distress are not managed well without clear domain adaptation [27]. This issue is enhanced in Nepali setting: the relationship between English social media consumption and the real mental health picture is not described well, and because of this, models trained on Western Reddit posts have certain implicit assumptions about the way distress is manifested that may not necessarily be transferable.

The field is more mature and the ethical issues have been brought into the limelight. Chancellor et al. [9] listed the list of risks: the problem with consent due to the necessity to analyze posts even without the direct consent, the problem of re-identification based on the allegedly anonymous text, the problem of stigmatization associated with unreasonable conclusions, the probability of harm in case the predictions are taken action without proper human supervision. Bender and Friedman [36] advocated obligatory data statements which record who created the data, under what circumstances and what has not been recorded. This is because these issues directly influenced the data collection and annotation steps outlined in Chapter 3.

This pre-trained language model is inherently biased in some ways as demonstrated below. The acknowledgment that mass pre-trained language models possess demographic biases has been turned into a single observation and nowadays assumed as one of the most critical NLP research issues. The underlying algorithm is not very complex: models get to know statistical patterns based on their training information, and training information is a reflection of the distribution as well as biases of the text that its writers generated. What has astonished researchers is the tenacity at which these patterns make it through fine-tuning and the extent to which they directly influence downstream predictions.

The first organized evidence was through the inert word embeddings. Bolukbasi et al. [21] demonstrated that the word2vec vectors would encode the gender stereotypes in a geometric manner, that a difference between the words man and woman gave a consistent gender direction in the embedding space, and occupation words fell along the same line in a pattern that was highly predictive of cultural stereotypes. This was not accidental; the geometry was understandable and regular. The same patterns were observed using glove embeddings [42], and further studies expanded the analysis to cover race, nationality, religion and age.

Upon BERT arrival, it was not evident whether contextual training would minimize these effects. It was soon demonstrated by counterfactual testing not to. Bhardwaj et al. [20] and later studies [22] were able to create sentence pairs that differed in demographic marker one male name was substituted by a feminine name, or a gendered pronoun one they were able to measure the change in the predictions by the model. Systematic shifts imply that the model is relying on demographic indicators, and not on the real content of the sentiment, which it

uses to make its prognostication. The changes were regular and significant to have an impact in practice.

Zhao et al. [23] made a significant addition: models do not only bring biases of the training data, they enhance them. The reason is that biased associations lower training loss and, thus, get more powerful gradient reinforcement in the course of training. A model that is trained on data in which, for example, some subreddits are weakly associated with some sentiment label, will have a stronger association than the data itself would indicate. This trend has been directly applied by Zhao et al. to the analysis of the demographic parity findings in Chapter 5 of this thesis as he termed this trend as bias amplification.

## **2.4 Bias in Pre-Trained Language Models**

The awareness that big pre-trained language models are demographically biased has become a widespread observation and is now a focus of NLP studies. The underlying process is not particularly complex: models are trained on statistical patterns on their training data, which is training data itself, which captures the distribution and biases of the text its authors have written. The question that has shocked scientists is the survival of these trends despite fine-tuning and the direct relationship to bottom-up predictions.

The earliest systematic evidence was the case of the word embeddings that were static. As Bolukbasi et al. [21] demonstrated, word2vec latencies were geometric binary representations of stereotypes of genders: The difference between "man" and "woman" vectors gave a fixed direction of gender in the encoding space, and occupation words were clustered along that direction in a manner that reflected cultural stereotypes well. This was not some peripheral result - the geometry could be interpreted and was consistent. The same patterns were observed in GloVe embeddings [42], and later it was further expanded to include race, nationality, religion, and age.

Contextual training was not evident when BERT came along as to whether these effects would be reduced. It was soon found counterfactual testing not to be. Bhardwaj et al. [20] and later research [22] created pairs of sentences identical excepting the demographic marker - the replacement of a male name with a female name or vice versa and gauged the extent to which the model changed in its predictions. Systematic changes suggest that the model is

operating on demographic signals instead of the sentiment content itself in coming up with its forecasts. The changes were frequent and big enough to be significant in reality.

The authors further contributed a significant point: models do not simply reflect biases of training data, they enhance them [23]. The way this works is that biased associations cause a loss of training to be minimized and hence reinforced more strongly during training. A model that is trained on the data with some subreddits having weak correlation with a sentiment label is going to be more strongly associated than the data would indicate. This trend was directly involved in the interpretation of the demographic parity results in Chapter 5 of this thesis, and is what Zhao et al. termed bias amplification.

The bias of dialect is especially worth being mentioned here due to direct impact of the qualities of fairness that are examined in this thesis. Blodgett et al. [27] demonstrated that sentiment classifying NLP systems do much worse on African American Vernacular English than on Standard American English. The difference is attributed to two issues which are related: non-standard varieties are not represented in pre-training corpora, and standard benchmarks lack subgroup evaluation, so the difference is not measured or reported. Other social media dialects have also been found to share the same findings [26]. In the case of mental health applications, the implication is that a model can be missing distress indicators that are written in non-standard English, not due to the distress being less real, but because the language pattern is less well known to the model.

Leavy [24] puts these technical findings in a larger argument: bias in NLP is not just a data problem. It is also an indication of who constructs such systems, which language they are you know, and which needs are considered as the norm. Technical remedies are solutions to symptoms but not causes. The same patterns were observed by Sap et al. [32], who observed patterns in social bias frames and models that were pre-trained on mainstream internet text failed to identify offensive or harmful statements directed towards specific groups when the phrasing was not similar to patterns in the training data. This type of failure is directly applicable in a mental health setting where most posts are characterized by stigmatizing social experiences.

The implication to this research is simple. Mental health can no longer tolerate systematic demographic differences in the manner that general-purpose systems may tolerate. A false

negative is not a statistical rounding error - it is an actual individual who was not marked. Such differences cannot be documented without demographically annotated data on evaluation, the vast majority of which are lacking in existing mental health datasets. It will necessitate running mitigation strategies in this particular domain, which was not the case before this study to determine whether they can be used to mitigate them.

## **2.5 Debiasing and Bias Mitigation Strategies**

The NLP fairness literature classifies the methods of mitigation by the point in the pipeline that they intervene: before training (pre-processing), during training (in-processing) or after training (post-processing). The trade-offs of each category are dissimilar with regard to flexibility, computational cost, and the extent to which they incur a change in the internal representations of the model.

Pre-processing procedures are applied to the training data and then it is presented to any model. One of the most popular is Counterfactual Data Augmentation (CDA) [21], [29]. The concept is to create pairs of training that are not different in terms of a demographic characteristic (i.e. male pronouns in the same sentence changed to female pronouns), and to make both of them part of the training set. When both versions are present in that model, the model will not be able to learn to use that marker as a predictor. Kaushik et al. [29] demonstrated better robustness on natural language inference tests and Huang et al. [35] demonstrated that counterfactual testing identifies failures modes that are absent in standard testing. The weakness of CDA is that it handles only those markers that you tell it to. Less obvious demographic cues, such as association with topics, word usage, style of writing, etc. are still present in the data and can be used. It also assumes that the swapped sentence will maintain the meaning to the word, which is really true when dealing with pronouns, but becomes a complex issue when dealing with names or terms that are culturally charged.

The in-processing approaches interfere with the process of training itself. The first method of this paper is adversarial debiasing [28]. It makes a primary classifier, in this case, a sentiment classifier, and an adversary, which attempts to extract demographic attributes based on the same learned representations. This is a gradient reversal layer between the encoder and the adversary: on backpropagation, a gradient reversal layer inverts the direction of the adversary gradients, such that the encoder should encourage harder demographic

prediction with the same sentiment error. The consequence is an encoder-decoder game that is a minimax game, that is, the encoder attempts to be both a good sentiment and useless at demographic inference. It was initially suggested by Ganin and Lempitsky [28] to adapt to the domain and has decreased gender-based difference in a variety of NLP problems. With respect to CDA, adversarial debiasing is not dependent on what demographic attributes to attack - the adversary can discover any signal the encoder is leaking. The disadvantage is that it is less stable than training: the minimax objective is sensitive to the learning rates and adversary capacity and convergence is not achieved.

Post-processing techniques do not modify the trained model and instead modify its threshold decision at inference time. Hardt et al. [30] demonstrated that the equalized odds that is, equal true positive and false positive results across groups can be obtained by a linear program resolution to obtain group-specific thresholds. This does not involve retraining or any special model, but does need group membership data at the time of inference, which is not always available, and does not do anything with the biased representations within the model.

One of the theoretical findings of Barocas et al. [37] provides significant background for interpreting the Chapter 5 findings. They demonstrated that in the presence of base rate differences between groups, demographic parity, equalized odds and calibration could not be met all three, in general. This is not a shortcoming of a particular technique - it is a mathematical one. Practitioners need to choose the criterion that is most important to them in their implementation and need to live with the understanding that their decisions often need to be made against one variable. This thesis analyzes all three and explains the trade-offs witnessed instead of talking about the measures that appear to be desirable.

Garg et al. [42] have also applied the bias analysis to historical embeddings, and demonstrated that word vectors trained on text data across decades follow changing gender and ethnic stereotypes through time - a result suggesting the extent to which social history is impregnated in these representations. In a broadly-read methodological criticism, Blodgett et al. [27] opined that viewing framing bias as an aspect of a technical issue complicated the social and political issue concerning whose language is treated as a standard and whose interests are met when designing the system. These two views influenced the design of RMH-Bias-10K and the presentation of the fairness analysis in this thesis. This was not only

aimed at improving the figures on fairness metrics, but also at identifying the remaining gaps and reporting them in a truthful manner.

## CHAPTER 3

### DATASET CONSTRUCTION

#### 3.1 Overview and Motivation

One of the challenges to research fairness in mental health NLP is the lack of demographically annotated data. The current corpora of mental health sentiment and risk classification algorithms, such as algorithms that were trained on the CLPsych shared tasks [16] or the UMD Reddit Suicidality Dataset [31], do not include any information about the demographics of the poster. This is not just a mere inconvenience. It is impossible to quantify whether a model will be differentially applied by different demographic groups without understanding which posts are posted by which group, or whether a certain group is more among those that are vulnerable to some form of bias, or whether a particular mitigation strategy results in real gains in fairness. The fairness issue reported on the literature cannot be empirically investigated on these corpora.

RMH-Bias-10K was constructed based on three realistic constraints. The quantity of dataset needed to be sufficient to aid model training and statistically significant fairness assessment across several demographic subgroups. It also needed to store both sentiment annotations and demographic metadata on multiple secured characteristics, as opposed to concentrating on one dimension of possible bias. Lastly, the annotation procedure needed to satisfy academic reliability criteria, the degree of formal inter-annotator agreement and independent expert validation was required, thus so the generated labels might be regarded as reliable ground truth.

Each step of the dataset construction process will be outlined below: platform and community selection, data collection and cleaning, sentiment annotation that was done in two stages, inter-annotator reliability analysis, demographic metadata generation, ethical considerations, and the final dataset structure.

Selection of platform and community: The choice of platform and community was influenced by the particular domain and selected population demographic, namely, healthcare professionals in this nation.

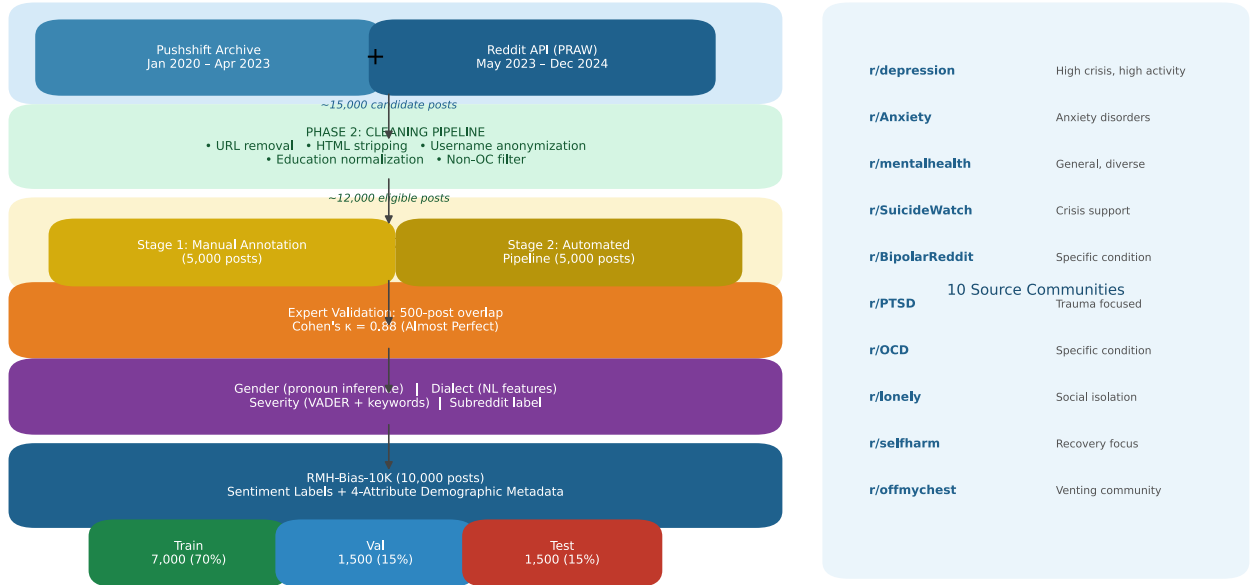


Figure 3.1: Dataset Construction Workflow showing the complete pipeline from raw Reddit data collection through cleaning, annotation, metadata generation, and final dataset assembly

### 3.2 Platform and Community Selection

The platform and community choice was based on the specific field and the chosen population demographic, i.e., the healthcare professionals in this country.

The choice of the source platform to use in RMH-Bias-10K was made because of a number of reasons. Mental health communities are some of the most active and largest on the internet on the platform, and extensive volumes of first-person mental health discourse are available. In comparison with microblogging sites like Twitter, where the amount of content shared in each post is limited by character censoring, posts on Reddit can be long and descriptive, containing more linguistic cues of the sentiment in question and aiding in demographic inference. The source has a proven history of being used in research in the field of mental health NLP [4], and the historical information could be obtained via the Pushshift archive [51], allowing gathering over a long-time frame.

Ten subreddits were sampled to belong to three criteria: the community should be focused on first-person experience of mental health conditions, not academic discussion, sharing of clinical information, or humor; the community should be active enough to provide a

sufficient amount of data; and the sub-reduction set as a whole had to cover a variety of conditions, the level of crisis severity, and the culture of the community to support meaningful subgroup analysis across the subreddit affiliation attribute.

Table 3.1: List of Subreddits Used for Data Collection

| Subreddit       | Community Focus                         | Activity Level |
|-----------------|-----------------------------------------|----------------|
| r/depression    | Depression experiences and peer support | High           |
| r/Anxiety       | Anxiety disorders and coping strategies | High           |
| r/mentalhealth  | General mental health discussion        | High           |
| r/SuicideWatch  | Crisis support and suicide prevention   | Medium-High    |
| r/BipolarReddit | Bipolar disorder lived experience       | Medium         |
| r/PTSD          | Post-traumatic stress disorder          | Medium         |
| r/OCD           | Obsessive-compulsive disorder           | Medium         |
| r/lonely        | Loneliness and social isolation         | High           |
| r/selfharm      | Self-harm and recovery journeys         | Medium         |
| r/offmychest    | Emotional venting and disclosure        | High           |

Even communities whose main activity is to collect news on mental health policy, disseminate clinical research, broadcast clinical diagnosis, and revolve around humor and entertainment were left out, although they talked about mental health issues. The first-person lived experience was the selection criterion, as it is in that context that the classification of sentiment is most significant, and the classification of demographics in its detection would have the most immediate implications on individuals.

### 3.3 Data Collection and Cleaning

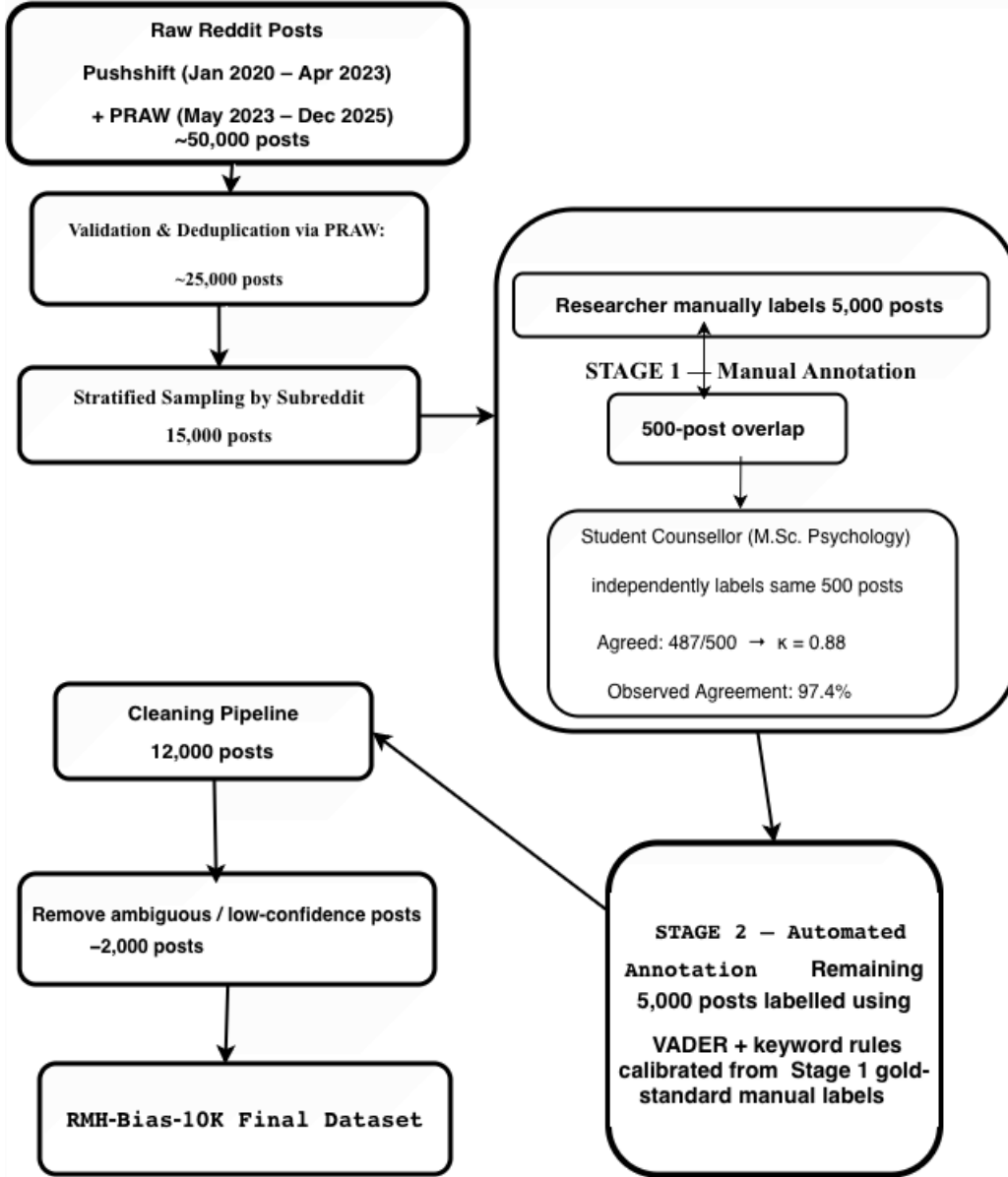


Fig. 3.1: Dataset Construction Workflow

### 3.3.1 Data Collection

The data were collected with two phases due to the alterations in the data access policy of Reddit throughout the time of study. During the initial step, which concerns the years 2020 to 2023, historical posts were retrieved with the help of the Pushshift Reddit archive [51]. Pushshift also made a full index of submissions and comments, and their full text, accessible, and not rate-limited, in bulk. The second phase took place in May 2023 to December 2024 when the official Reddit API was read via

Python Reddit API Wrapper (PRAW) after Pushshift was closed down and moved to a public endpoint.

The filtering criteria used in both phases were the same. Submissions made by oneself were only retained, and link posts that redirected to an external post were excluded. Initial cleaning had to be of at least 50 characters to guarantee something meaningful of sentiments. Some posts were removed, either because they were highlighted as deleted, removed by the moderators or because they were authored by accounts that were classified as a bot (through pattern of usernames and comment history). Only English language contents were collected, and they were identified using the langdetect library with the confidence threshold set to 0.95.

Follow-up PRAW was also applied following the previous retrieval to confirm the current positions of all posts and eliminate those that were deleted or removed in the meantime. A stratified sampling based on subreddit was then carried out to form a diverse candidate pool of about 15,000 posts to be used in the cleaning and annotation pipeline, with each sample community being equally represented.

### **3.3.2 Cleaning Pipeline**

The systematically curated pipelines were applied to each candidate post that was filtered to eliminate technical artifacts without destroying linguistically useful information with semantic import and demographic indicators. The following operations were used.

In the initial step, URLs were eliminated using a regular expression of the patterns of HTTP and HTTPS as hyperlinks do not include a sentiment signal. BeautifulSoup was then used to clear HTML tags and API-encoded entities. The mentions of the usernames in the u/username format were removed to maintain the anonymity and to minimize the token noise that may disrupt the downstream classification. In addition to these structural artifacts, Markdown formatting characters, emphasis asterisks, headings hashes, code backticks were removed and the underlying text remained intact. The repetitive uses of punctuations that included runs of marks of exclamation or ellipses more than three periods were crushed into one. Lastly, the whitespace

leading and trailing was eliminated and the run of internal whitespace was trimmed to one space.

After cleaning, two groups of posts were ruled out. The posts where over 30 percent of the rest of the material was post-quoted text (defined by the standard quoting markup used on Reddit) were deleted since they are not an expression but a response and cannot be safely attributed to the sentiment of the author. Posts where the majority of the content was made up of symptom checklists, medication schedules, diagnostic criteria, or other structured clinical data were also filtered, as these are informational as opposed to expressive and that the personal sentiment is not analyzable.

Following the cleaning and exclusions, the annotation-eligible pool had approximately 12,000 posts, or a reduction of only about 20 per cent of the initial 15,000-posts candidate set.

### **3.4 Sentiment Annotation**

Sentiment annotation was carried out in two steps. Stage 1 involved manual labeling of a stratified subsample and an independent expert annotator was used to validate the results. Stage 2 amplified coverage of the rest of the posts using an automated pipeline which was calibrated using Stage 1 gold-standard labels. This design provided a tradeoff between the quality of annotations and scale with the best labels being found in the manually annotated subset that was to be used in the calibration and validation exercise.

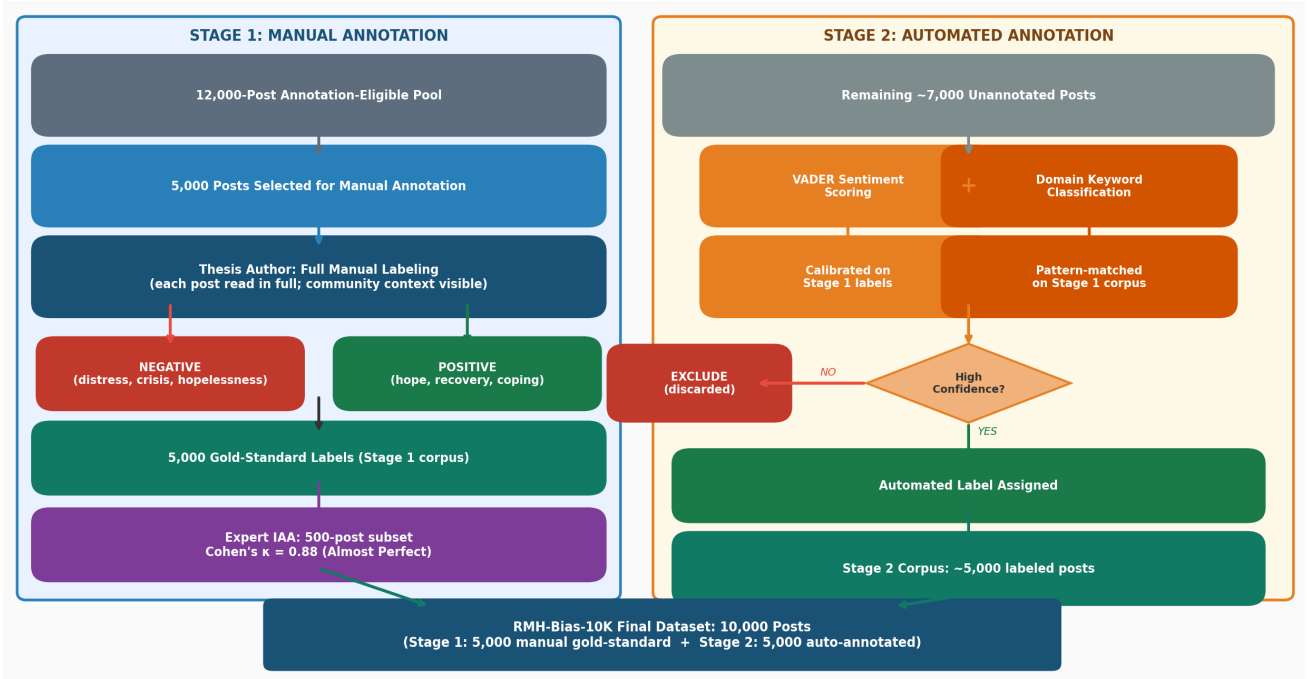


Fig. 3.2: Two-Stage Annotation Pipeline showing Stage 1 manual annotation with expert validation and Stage 2 automated annotation with confidence filtering

### 3.4.1 Annotation Schema

Sentiment was a binary label on the level of the post. The Negative category involved posts about distress, hopelessness, suicidal ideation, self-harm, acute psychiatric symptoms, a request of crisis support, or an important decline in mental health. The Positive category consisted of those posts which stated hope, recovery advancement, thankfulness of support, encouragement of others, active coping or meaningful mental health enhancement. True neutral posts, i.e., factual questions on medications or treatments, information requests, or posts whose tone cannot be identified but have none, were not included in the dataset but instead left out. This separation ensured a clean division of classes and noisy labels were prevented.

In the case of mixed emotions posts, the overall emotional inclination of the whole post was applied instead of identifying solitary sentences. Dark humor or irony to express underlying distress was called by the name and number of the actual emotional state as opposed to the tone of the surface, as in standard mental health annotation practices. Regular use of these rules meant that one had to be familiar

with the conventions of mental health discourse on Reddit, hence the need to have expert validation as a significant component of the procedure.

### **3.4.2 Stage 1: Manual Annotation**

Stage 1 was the manual labelling of 5,000 posts selected by sampling among 12,000-post pool of posts that were eligible to be annotated so that each of the ten subreddits would have comparable representation. All the posts were read and their subreddit of origin was revealed, as the community rules impact on the interpretation of the emotional expression. There were no automated instruments or external resources and the aim was to produce human judgments without aids and that it can be used as a good guide to be calibrated later. This took a number of weeks.

The borderline cases were recorded in a systematic manner and the annotation rules were honed down. Dark-humor posts (where ironic surface language concealed real distress) and mixed-register posts (when suffering was present together with recovery) and community-norm posts (when the standard expressions of support were used, but they did not reflect the actual condition of the author) were the most difficult types. The basing of decisions was done based on the general post context and community norms, and the reasoning was documented to ensure consistency.

The third recommendation states that the independent expert validation should be conducted independently.

### **3.4.3 Independent Expert Validation**

A Master student in Psychology who is a practicing student counselor was then hired as an independent annotator to validate Stage 1 labels and add to the clinical expertise. This person introduced expertise in mental health evaluations and crisis specialists.

The professional coded one stratified random sample of 500 posts of the Stage 1 sample, stratified by subreddit and sentiment tag. Only the written schema and operational definitions were provided to the annotator, who had no choices on the original labels.

### **3.4.4 Stage 2: Automated Annotation**

Stage 2 automatified the rest of the posts, using the labels of the gold standard of Stage 1 as a calibration. The pipeline used a domain-specific key element based on the patterns observed in the process of manually annotating patterns as well as VADER compound sentiment scores [8]. VADER scores lie between the negative and positive value of -1 to +1 respectively. The keyword term encompassed both positive words (recovery, progress, gratitude, hope) and negative words (crisis, hopelessness, suicidal ideation, acute distress) that were characterized in the Stage 1.

Calibration to thresholds and weights of the key words were performed on a held-out subset of Stage 1 data to maximize consistency with the gold-standard labels. The posts that generate low-confidence or conflicting outputs were filtered out. Low confidence was established where VADER scores were -0.05 to +0.05 where VADER reliability is known to decline. The indication of the contradictory results was the VADER and keyword components that indicated same directions and no clear dominance in the direction.

The final RMH-Bias-10K corpus was created after Stage 2 annotation and removed any post that was deemed uncertain.

### **3.5 Annotation Reliability**

#### **3.5.1 Inter-Annotator Agreement Calculation**

The inter-annotator agreement is calculated by the number of documents that the respondent agreed with divided by overall response to the questions. Concordance between the author of the thesis and the expert annotator was calculated on the set of 500 posts overlapping. Out of 500 posts, 487 were labeled as such by both annotators thus yielding a 97.4% agreement with 13 disagreements. Raw agreement is not a reliable test used in cases where there is an uneven distribution of classes since a naive classifier can always be guaranteed to be in the majority class thus giving high observed agreement by chance. Cohen kappa is a chance-corrected statistic which takes into account the expected agreement when assuming that the marginal class distributions are independent random labeling.

Cohen's kappa is defined as:

$$K = (Po - Pe)/(1 - Pe) \quad (\text{Eq. 3.1})$$

where  $Po$  is the observed proportional agreement and  $Pe$  is the expected agreement by chance.  $Pe$  is calculated from the marginal label frequencies as:

$$Pe = (pA_{neg} \times pB_{neg}) + (pA_{pos} \times pB_{pos}) \quad (\text{Eq. 3.2})$$

Using the label distributions observed in the 500-post overlap set, where 88.1% of posts were labeled Negative and 11.9% Positive, the expected chance agreement is:

$$Pe = (0.881 \times 0.881) + (0.119 \times 0.119) = 0.776 + 0.014 = 0.790$$

Substituting into the kappa formula:

$$K = (0.974 - 0.790) / (1 - 0.790) = 0.184 / 0.210 = 0.88$$

Table 3.2: Inter-Annotator Agreement Results

| Metric                             | Value                                       |
|------------------------------------|---------------------------------------------|
| Annotator 1                        | Thesis Author (Computer Information System) |
| Annotator 2                        | Student Counsellor (M.Sc. Psychology)       |
| Posts in overlap set               | 500                                         |
| Agreed posts                       | 487                                         |
| Disagreed posts                    | 13                                          |
| Observed Agreement ( $Po$ )        | 97.4%                                       |
| Expected Chance Agreement ( $Pe$ ) | 79.0%                                       |
| Cohen's Kappa ( $K$ )              | <b>0.88</b>                                 |
| Kappa Interpretation               | Almost Perfect Agreement [44]               |

### 3.5.2 Interpretation of Agreement

The interpretation of the Agreement is subject to all of these. The kappa of 0.88 fits in the almost perfect agreement range based on standard thresholds [44]. This is ranked as being powerful in mental health annotation which is a field characterized

by complexity in interpretation, emotional uncertainties and language which is culturally based. This finding is further reinforced by the finding that the result was obtained between computational and clinical annotator groups, which gives more credence to the validity of labels than the agreement when annotators are of a similar training environment.

### **3.5.3 Disagreement Analysis and Resolution**

A reconciliation session was scheduled to discuss the 13 posts that disagreed following independent labeling. Through analysis, two primary recurring patterns were found: dark-humor posts wherein sarcasm or self-deprecating irony produced a positive surface tone and but underlying distress; and mixed-coping posts wherein apparent acute distress and active recovery existed together in the same text and the predominant orientation was indeed ambiguous. All cases were discussed in a structured way and consensus labels were arrived at, as well as the rationale was recorded to improve on the guidelines used in Stage 2.

### **3.6 Bias Metadata Layer**

Since the demographic attributes of the RMH-Bias-10K are not self-reported in a structured format by the users of Reddit, and the re-identification of the posts with the identity of the individual user would present a significant ethical issue, all four demographic attributes of the dataset were estimated using the textual content of the post at the post level, not the user-level. These are proxy labels and not validated ground-truth demographic measurements and these implications on the reliability of fairness evaluation are discussed in Section 3.7.

A combination of self-referential patterns of pronoun, gendered pattern of relational term and analysis of username patterns were used to infer gender. Articles with regular use of he/him and homonyms, use of gender relational words like my boyfriend, my father, my brother or usernames with examples of typical man names were designated as male-inferred. Posts that used she/her, gendered relationship words, like my girlfriend, my mother, or my sister, or were named after a female were identified as female-inferred. The posts where no gender cues, where gender cues were unclear or contradictory were placed into an unknown category and were not subjected to gender-based fairness analysis. About 35 percent of the posts contained enough clear and consistent gender indicators to be given binary category,

and 65 percent of the rest were placed in the unknown category. The binary system fails to represent non-binary or transgender identities, which is also known to be a limitation.

Dialect was termed either as Standard or Non-Standard English, depending on whether the dialect had one or more recognizable non-standard linguistic features and the extent. Characteristics applied to define Non-Standard English were grammatical patterns aligned with African American Vernacular English, like habitual be construction, copula deletion and aspectual done; vocabulary items, which are regional, that do not appear in Standard American English, and systemic non-standard spelling patterns, which are repeated and consistent throughout a post, as opposed to the occurrence of individual typographical mistakes. The posts that contain many such features in comparison with the total number of words were deemed non-standard; others were deemed Standard. This dual discrimination dramatically simplifies the real scale and complexity of dialectal variation, yet it brings into discernible measure the scale of linguistic variation that is directly connected with the prejudice recorded in the literature [27].

Table 3.3: Bias Metadata Distribution in RMH-Bias-10K

| Attribute       | Category             | Count | Percentage |
|-----------------|----------------------|-------|------------|
| Gender          | Male-inferred        | 1,850 | 18.5%      |
|                 | Female-inferred      | 1,680 | 16.8%      |
|                 | Unknown              | 6,470 | 64.7%      |
| Dialect         | Standard English     | 7,420 | 74.2%      |
|                 | Non-Standard English | 2,580 | 25.8%      |
| Crisis Severity | High Severity        | 3,240 | 32.4%      |
|                 | Low Severity         | 6,760 | 67.6%      |
| Sentiment Label | Negative             | 8,810 | 88.1%      |
|                 | Positive             | 1,190 | 11.9%      |

The severity of the crisis was determined based on a hybrid of VADER compound sentiment scoring [8] and keyword matching in domains. High Severity Posts in this category were those scoring strongly negative in VADER (a score below negative 0.5), and containing direct language of crisis, including single statements of suicidal thoughts, active self-harm intentions, acute psychological breakdown, or statements about not being able to

go on. The classification of posts was based on Low Severity when it was characterized by continuing difficult life condition, a low mood, or a state of overall distress without signs of acute crisis. This method of measuring severity is an approximation measure of clinical assessment which would otherwise involve a structured interview and trained judgment. The proxy represents a significant difference in the urgency and distress severity but it must not be compared to clinical diagnosis or risk evaluation.

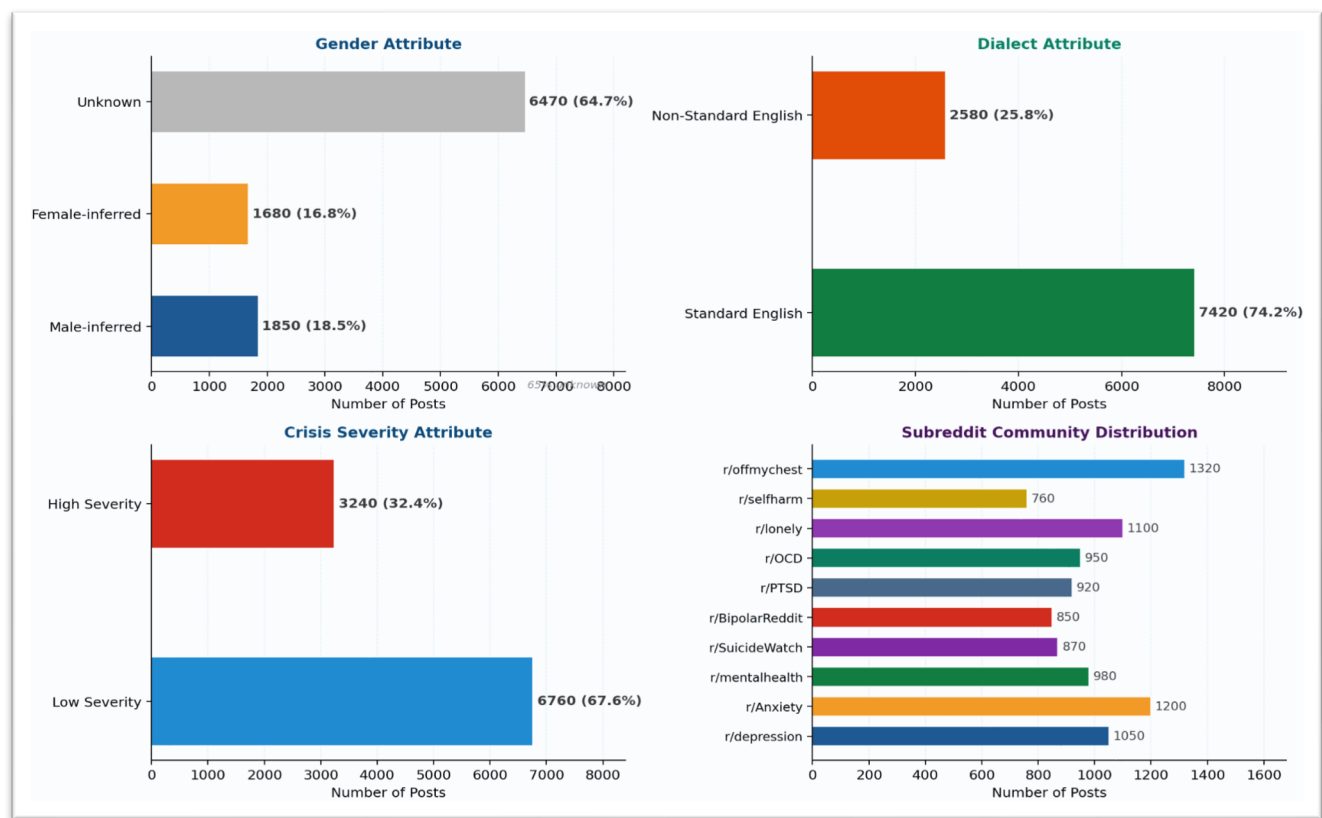


Figure 3.3: Demographic Attribute Distributions in RMH-Bias-10K showing bar charts for gender, dialect, crisis severity, and sentiment label across the 10,000-post dataset

### 3.7 Ethical Considerations

All the posts in RMH-Bias-10K are completely anonymous. Timestamps were recorded accurately to allow individual identification and usernames, post identifiers, direct hyperlinks, and timestamps were not included in the final dataset. The post material that is used as an example in this thesis has been copied into a composite form that maintains the

interested linguistic features without making the given text searchable. Such a practice would not allow tracing the examples to its authors.

The Terms of Service of Reddit allow the collection of data under the non-commercial research. Each of the posts gathered was posted willingly to publicly available subreddits where the content was not restricted by membership and privacy settings. The study was conducted in the ethical principles of social media research in mental health setting by De Choudhury et al. [4], which holds that the research needs to rely on publicly available data, that the analysis needs to be done at the aggregate, not on a case-by-case basis, as much as possible, and the research must not be in a way that could cause harm to the participants. These guidelines guided every phase of data gathering, annotation, and presentation.

The deduced demographic tags incorporate a definite ethical factor that must be taken into account. Categorizing a post as female-inferred when the writer did not explicitly state that she was female, or non-standard dialect when the writer might be a second language English speaker and not an English speaker of a given variety of a certain dialect, will be categorizing people in a way that they did not identify with. This risk is understood and one of the reasons that the labels are always defined as inferred proxies and not confirmed demographic facts all through the thesis. The fairness test which relies on such labels is construed correctly as demonstrating systematized model behavioral patterns with respect to language and textual characteristics that are associated with demographic groups, as opposed to a final declaration over how particular individuals are subject to bias.

Uncondensed post information does not get redistributed publicly. The metadata (demographic) is kept in encrypted institutional storage with limited access with the minimum identifiability required to use it in research. The study was carried out within the regulations of academic integrity and ethical research of Nepal college of information technology and Pokhara University.

### **3.8 Final Dataset Structure**

The final RMH-Bias-10K dataset is a 10,000 posts dataset which has been stratified randomly into three groups: training, validation and test. The stratification was used in combination with sentiment label, subreddit of origin and gender category in order to

guarantee the proportional representation of each combination of attributes across all the three splits. The split was only done once and was fixed in all experiments reported in this thesis, to ensure that all model comparisons are performed on the same data partitions, they are not confounded by a change in data splitting.

Table 3.4: Dataset Statistics by Split

| <b>Split</b> | <b>Total Posts</b> | <b>Negative</b> | <b>Positive</b> | <b>Negative %</b> |
|--------------|--------------------|-----------------|-----------------|-------------------|
| Training     | 7,000              | 6,167           | 833             | 88.1%             |
| Validation   | 1,500              | 1,322           | 178             | 88.1%             |
| Test         | 1,500              | 1,321           | 179             | 88.1%             |
| <b>Total</b> | <b>10,000</b>      | <b>8,810</b>    | <b>1,190</b>    | <b>88.1%</b>      |

Each record in the final dataset contains the following fields: cleaned post text, binary sentiment label (0 for Negative, 1 for Positive), subreddit of origin, inferred gender category (male, female, or unknown), dialect type (Standard or Non-Standard), crisis severity level (High or Low), approximate word count, and the posting year. The 88.1% to 11.9% class ratio, which is preserved across all three splits, reflects the genuine distribution of positive and negative sentiment expression in the source mental health communities and represents a meaningful challenge for model training that is addressed through the class imbalance handling strategies described in Chapter 4.

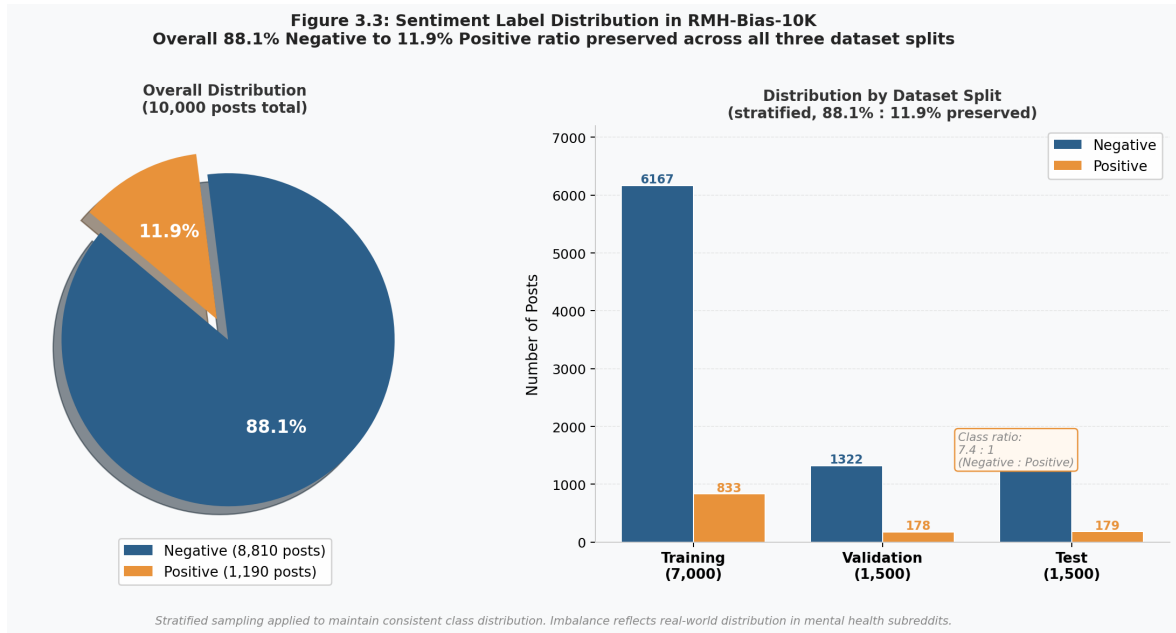


Figure 3.4: Sentiment Label Distribution in RMH-Bias-10K showing the 88.1% to 11.9% negative-to-positive split across training, validation, and test sets

### 3.9 Comparison with Other Datasets

Table 3.5: Comparison of RMH-Bias-10K with Existing Mental Health NLP Datasets

| Dataset                   | Size            | Task                      | Demographic Annotations              | Platform |
|---------------------------|-----------------|---------------------------|--------------------------------------|----------|
| CLPsych 2019 [16]         | ~500 users      | Suicide risk (4 levels)   | None                                 | Reddit   |
| UMD Suicidality [31]      | ~10,000 users   | User-level classification | None                                 | Reddit   |
| GoEmotions [17]           | 58,000 comments | 27-class emotion          | None                                 | Reddit   |
| RMH-Bias-10K (this study) | 10,000 posts    | Binary sentiment          | Gender, dialect, severity, subreddit | Reddit   |

The closest existing resources are The CLPsych 2019 dataset and the UMD Reddit Suicidality Dataset. Both are annotated, careful, and Reddit-based mental health corpora, but neither contains demographic metadata of any sort. The fairness assessment performed in

this thesis cannot be reproduced on either corpus without demographic metadata of any kind. GoEmotions is more linguistically diverse and larger, but focuses on establishing emotional categorization in a broad field, and not mental health in particular, and also is not annotated demographically. RMH-Bias-10K bridges this particular gap by incorporating multi-dimensional demographic metadata and a domain focused mental health corpus.

The trade-offs that are considered in the RMH-Bias-10K design must be identified explicitly. The binary sentiment schema is not as clinical as the four-level risk classification in CLPsych, which restricts the kind of research questions that can be supported by the dataset. The demographic labels are not self-reported but derived, which brings in noise in measurements. The scope is limited to the English-language content in Reddit, limiting its generalizability. These weaknesses do not affect the main value of the dataset, which is to allow systematic assessment of fairness in this field of study first, but must be taken into account in the interpretation and extrapolation of findings on the basis of the dataset.

## CHAPTER 4

### METHODOLOGY

#### 4.1 Overview

This chapter describes the experimental structure that is set to test three model architectures with the help of the RMH-Bias-10K dataset. The main idea of this methodology is the concept of controlled comparison; by keeping training data, evaluation protocols, hardware and hyperparameter tuning constant, one can isolate any performance difference. Therefore, the observed variations can be explained by the distinct model structures, or by the inclusion of fairness interventions, but not by accidental variations in the work of the experiment.

The models that were evaluated include M1, the BiLSTM model based on GloVe embeddings and the attention mechanism to act as a traditional deep learning baseline; M2, the vanilla BERT model fine-tuned on binary sentiment classification, which is the transformer baseline; and M3 (FairBERT). The latter is a generalization of M2 which combines model-level adversarial debiasing with data-level counterfactual augmentation. To provide the strength of these results, five experimental runs of each model were done with different random seeds (42, 123, 456, 789 and 1024). This multi-seed design, or a total of fifteen different runs, allows one to compute means and standard deviations, having the required statistical background of significance testing.

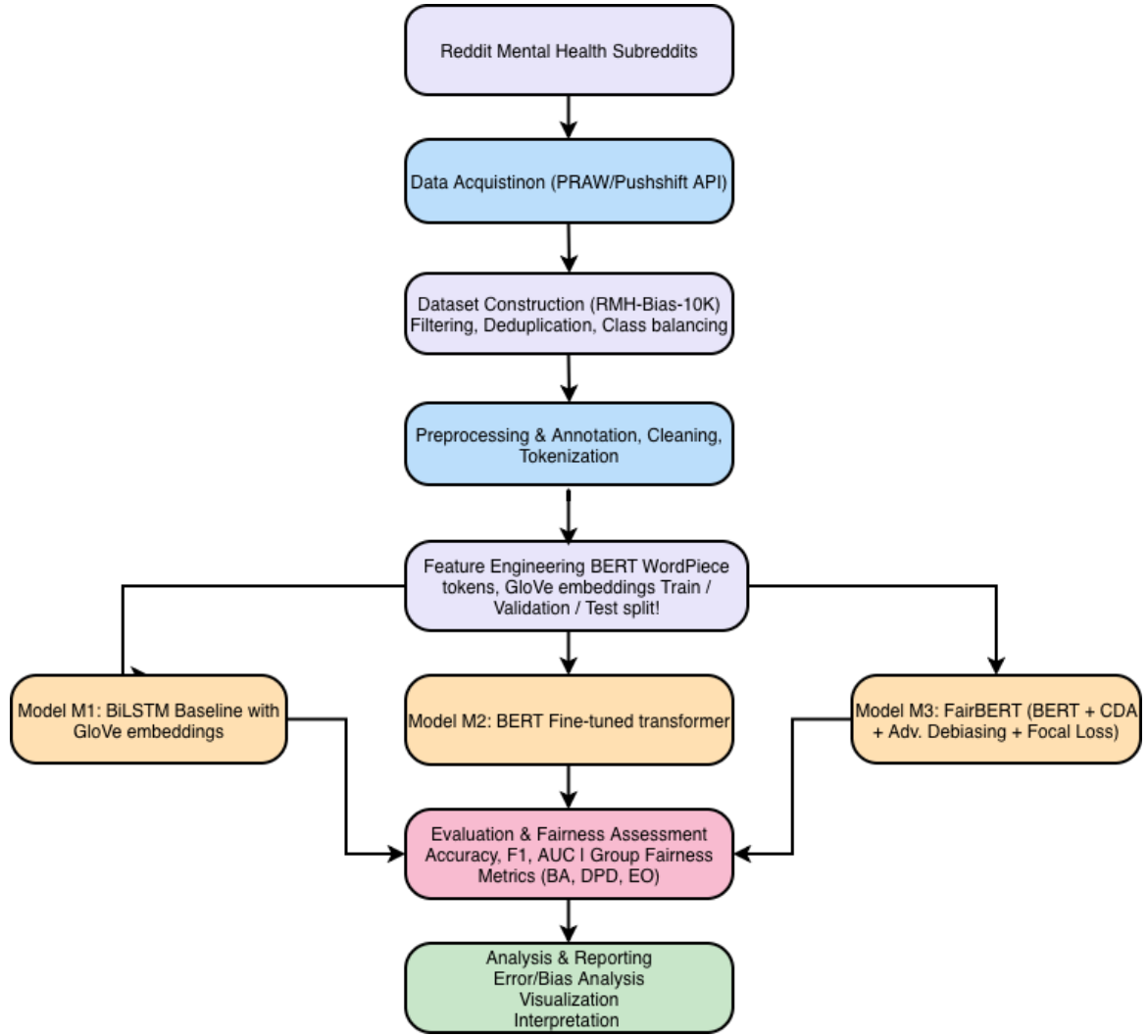


Fig. 4.1: Overall Methodological Workflow for all three models

## 4.2 Research Design

The experimental design is in the form of a controlled within-subject comparison with the model architecture as a primary independent variable and the fairness intervention as a secondary one. The study is effective in isolating the specific effect of these two factors by holding all other factors constant. Moreover, statistical inference has a strict basis of the choice to replicate the seeds with five seeds. Instead of considering the performance of a model as a deterministic value of a single training dynamics, this method considers results as a sample of a wider range of possible outcomes of randomization. As a result, paired t-

tests performed on these seeds show whether any observed performance deltas are actual changes or of a chance nature.

The dependent variables will be in two categories. The performance dependent variables are accuracy, macro-averaged precision, macro-averaged recall, macro-averaged F1-score, and AUC-ROC which are calculated on the held-out test set. Dependent variables associated with fairness encompass demographic parity difference (DPD), equalized odds gap (EDO) and accuracy parity gap (BA), which are calculated separately in respect to the four protected attributes, which are: gender, dialect, crisis severity, and subreddit community. This is why such joint assessment in both groups is the defining feature of the current study in comparison to most of the other currently existing mental health NLP studies where only aggregate performance is provided.



Fig 4.2: Overall Experimental Framework showing the three-model comparison design with shared data pipeline, independent training with five seeds each, and joint performance and fairness evaluation

Comparisons between seeds within a model determine the stability and determine sensitivity to random initialization. Between-model studies determine the impact of architectural

selection on performance and equity. For example, the M2 vs M3 comparison is of particular interest as it is the only model that isolates the effect of the fairness interventions: both models are based on the same underlying BERT architecture and aim to achieve the same objective which is fine-tuning, but M3 additionally results in adversarial debiasing and counterfactual augmentation.

## **4.3 Data Preparation**

RMH-Bias-10K data set is preprocessed systematically to provide consistency in all experimental settings with linguistic characteristics when applied to sentiment analyzing in the context of mental health.

### **4.3.1 Text Cleaning**

The post cleaning of raw Reddit posts contains various specific transformations. Text is all lowercased in order to control the size of vocabulary and enable more effective model generalization. To reduce non informative noise, URLs are removed since URLs seldom have sentimental meaning and the HTTP/HTTPS links together with the rest of the HTML tags that appear after scraping are eliminated. As the repetition of punctuations (e.g. “!!!”) is reduced to one case, certain emotional indicators are maintained to retain the intensity of the post made by the user. Lastly, the text is made standardized through a reduction of more than one space, tab, and newlines to one space.

This pipeline aims at developing a compromising balance between reducing noise and information loss. This method is more grammatical and retains grammatical constructions and emotive intensifiers, such as very, really, and extremely, in contrast to more aggressive preprocessing methods, such as stopword removal or stemming. These features are a crucial indication of sentiment classification that would have been overlooked [41].

### **4.3.2 Counterfactual Data Augmentation**

In the case of the M3 model alone, training set is expanded using counterfactual examples that have been created by systematic gender-term substitution. The task is to generate artificial examples of training in which the markers of gender are exchanged, and the main semantic meaning is retained [29, 35]. This can be performed in the following steps: This procedure consists of the following steps: first, identifying certain words in the original text, replacing them with output to form a counterfactual result and then reintroducing these results into the augmented training set.

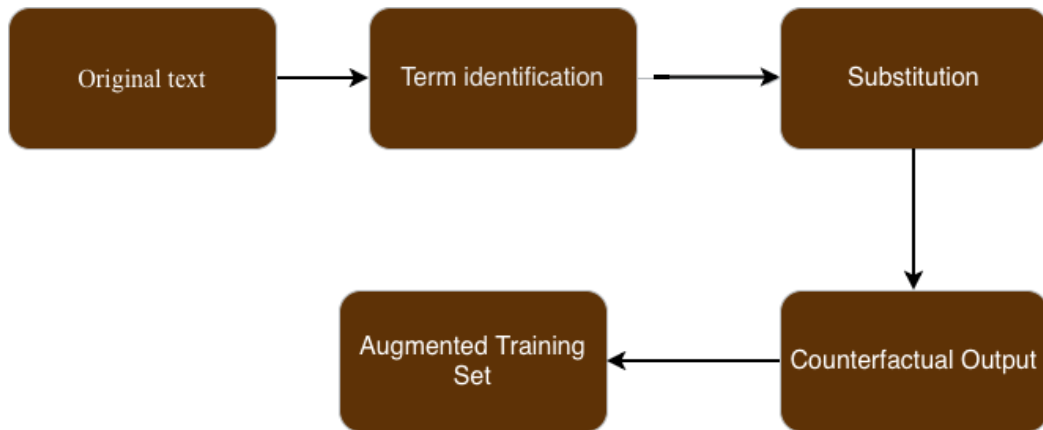


Fig. 4.3: Counterfactual data augmentation process pipeline.

Gender Swap Pairs:

*he ↔ she | him ↔ her | his ↔ hers | man ↔ woman | boy ↔ girl*  
*boyfriend ↔ girlfriend | husband ↔ wife | father ↔ mother*  
*brother ↔ sister | son ↔ daughter*

Listing 4.1: CDA Implementation

```

import re
GENDER_SWAPS = {'he': 'she', 'him': 'her', 'man': 'woman', ...}
def apply_cda(text):
    for src, dst in GENDER_SWAPS.items():
        text = re.sub
        (rf'\b{src}\b', dst, text, flags=re.IGNORECASE)
    return text
  
```

The substitution uses word boundary constraints (regex `\b` anchors) to prevent partial replacements inside longer words (for example, replacing “her” inside “therapy”). Each post that contains at least one gender marker generates one counterfactual variant, which results in an augmented training set roughly 1.8 times the original size. Importantly, augmentation is applied only to the training data. The validation and test sets keep their original form so that fairness metrics reflect real distributional properties rather than artificially balanced data [42].

### 4.3.3 Label Encoding and Data Splitting

Sentiment labels were encoded as binary integers: Negative = 0 and Positive = 1. The dataset was split into training (70%), validation (15%), and test (15%) partitions using stratified random sampling to preserve the 88.1% to 11.9% class distribution across all three partitions. The split was determined once with a fixed random seed and applied identically to all three models and all five training seeds. Using the same fixed splits everywhere ensures that differences in performance and fairness metrics come from model behavior, not from variation in data partitioning.

Table 4.1: Stratified Dataset Split

| <b>Split</b> | <b>Size</b> | <b>Percentage</b> | <b>Negative</b> | <b>Positive</b> |
|--------------|-------------|-------------------|-----------------|-----------------|
| Training     | 7,000       | 70%               | 6,166           | 834             |
| Validation   | 1,500       | 15%               | 1,322           | 178             |
| Test         | 1,500       | 15%               | 1,322           | 178             |
| <b>Total</b> | 10,000      | 100%              | 8,810           | 1,190           |

## 4.4 Handling Class Imbalance

The 88.1% to 11.9% class split in RMH-Bias-10K is a genuine training problem. A classifier that predicts the majority label for every single input achieves 88.1% accuracy without learning anything useful -- that is not a model, it is a default. Neural classifiers left unchecked drift toward exactly this behavior because majority-class predictions consistently produce lower training loss, and the minority-class boundary never gets enough gradient signal to

form properly. The result looks acceptable in aggregate but falls apart the moment you look at positive-class performance specifically. That matters here because positive-class posts, people writing about recovery, coping, or moments of stability, are not noise to be tolerated. Misclassifying them consistently is a fairness failure as much as a performance one, and accepting it for the sake of a cleaner overall accuracy number would undermine the whole point of the study.

Table 4.2: Class Imbalance Handling Strategies\\

| Model | Under sampling | CDA | Weighted Loss | Loss Function                                              |
|-------|----------------|-----|---------------|------------------------------------------------------------|
| M1    | Yes            | No  | Yes           | Weighted Cross-Entropy                                     |
| M2    | Yes            | No  | Yes           | Weighted Cross-Entropy                                     |
| M3    | Yes            | Yes | Yes           | Focal Loss ((gamma) $\gamma=2.0$ , (alpha) $\alpha=0.25$ ) |

Two countermeasures are shared across all three models. First, random of the negative class brings the training batch ratio down to roughly 4:1 negative-to-positive. This does not eliminate the majority class signal – doing so would discard genuine information – but it reduces how aggressively the majority class dominates each gradient update. Second, class-weighted loss assigns the positive class a weight of approximately 7.4, the inverse of its frequency in the training set. This directly increases the penalty for misclassifying a positive example, forcing the model to treat minority-class errors as more costly than the raw loss function would suggest on its own.

For a training example with predicted probability  $p$ , the focal loss is:

$$FL(p) = -\alpha \times (1 - p)^{\gamma \log(p)} \quad (\text{Eq. 4.0})$$

Here  $\gamma$  controls the focusing strength (set to 2.0 in M3) and  $\alpha$  balances the class weighting (set to 0.25). With  $\gamma = 2.0$ , the loss from examples the model already classifies confidently is reduced by up to a factor of four relative to hard examples. This focusing effect is particularly helpful in imbalanced settings where the model can quickly learn to classify the majority class with high confidence and then plateau, without devoting enough capacity to learning the minority class boundary.

M3 goes further by swapping standard cross-entropy for focal loss [38] on the sentiment classification objective. The core idea is simple: once a model has learned to classify easy examples confidently, those examples stop teaching it much – their loss is already low, so their gradient contribution is small. Standard cross-entropy does not account for this and keeps weighting easy and hard examples by the same principle. Focal loss adds a modulating factor that explicitly scales down the contribution from well-classified examples and redirects training capacity toward the harder, ambiguous ones the model is still getting wrong. The formulation is given in Eq. 4.0, where gamma controls the focusing strength and alpha handles class weighting. In M3, gamma is set to 2.0 and alpha to 0.25. At gamma = 2.0, the loss from a confidently classified example is reduced by up to a factor of four relative to a hard one. In an imbalanced setting like this, where the model can reach a comfortable plateau by learning the majority class boundary early and then coasting, that focusing pressure is what keeps training from stalling before the minority class is adequately learned.

## **4.5 Model Architectures**

This section provides detailed descriptions of the three model architectures evaluated in this study. Each architecture represents a distinct approach to sequence modeling and sentiment classification, progressing from recurrent networks (M1) to pre-trained transformers (M2) to fairness-augmented transformers (M3).

### **4.5.1 M1: BiLSTM with Attention Mechanism**

The BiLSTM model processes text as sequences of word embeddings, serving as a recurrent neural network baseline that predates transformers.

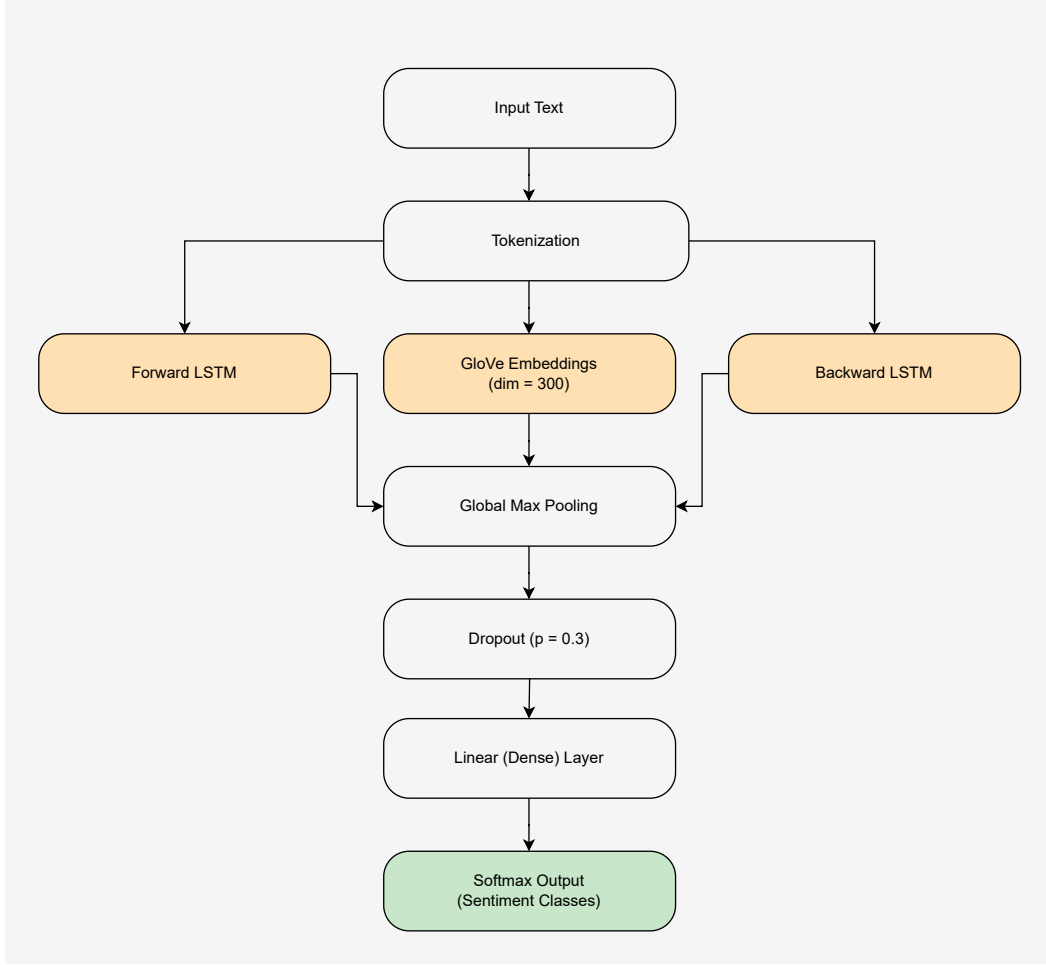


Fig. 4.4: BiLSTM with GloVe embedding architecture.

The architecture has four main components. First, the embedding layer maps words to 300-dimensional GloVe vectors [11]. The embedding matrix is frozen during training to take advantage of semantic knowledge learned from a large-scale corpus (6 billion tokens, 400K vocabulary). Second, two stacked bidirectional LSTM layers process these embeddings in both forward and backward directions, producing hidden states  $h_t$  of dimension 256 at each position. Each LSTM cell uses standard gating mechanisms [10] to handle long-range dependencies. Third, an attention mechanism learns to weight each position by its relevance to the classification task [44]:

$$attention\_weights = softmax(\tan h(W_a \times h_t)) \quad (\text{Eq. 4.1})$$

$$context_{vector} = \sum (attention_{weights} \times h_t) \quad (\text{Eq. 4.2})$$

Fourth, the context vector passes through dropout (p=0.3) and a fully connected layer to produce binary sentiment predictions.

Table 4.3: M1 BiLSTM Hyperparameters

| Hyperparameter                   | Value                    |
|----------------------------------|--------------------------|
| Word embedding dimension         | 300 (GloVe, frozen)      |
| LSTM hidden size (per direction) | 128                      |
| LSTM layers                      | 2 (stacked)              |
| Maximum sequence length          | 256 tokens               |
| Dropout rate                     | 0.3                      |
| Optimizer                        | Adam                     |
| Learning rate                    | 0.001                    |
| Batch size                       | 32                       |
| Gradient clipping                | Max norm 1.0             |
| Early stopping patience          | 3 epochs (val. macro F1) |

#### 4.5.2 M2: Vanilla BERT for Sentiment Classification

M2 fine-tunes bert-base-uncased (110 million parameters, 12 transformer layers) for binary sentiment classification without any fairness-specific modifications.

BERT uses masked language modeling pre-trained on large text corpora [14], producing contextual representations that can be adapted to downstream tasks through fine-tuning.

The input representation works as follows. Text is tokenized into WordPiece subword units [45], and each sequence begins with a [CLS] token and ends with a [SEP] token. The 12 transformer layers of multi-head self-attention and feed-forward networks process this input, producing contextual representations at each position. Each layer uses residual connections and layer normalization [46]. For classification, the final hidden state of the [CLS] token serves as the sequence representation and is projected to binary sentiment logits.

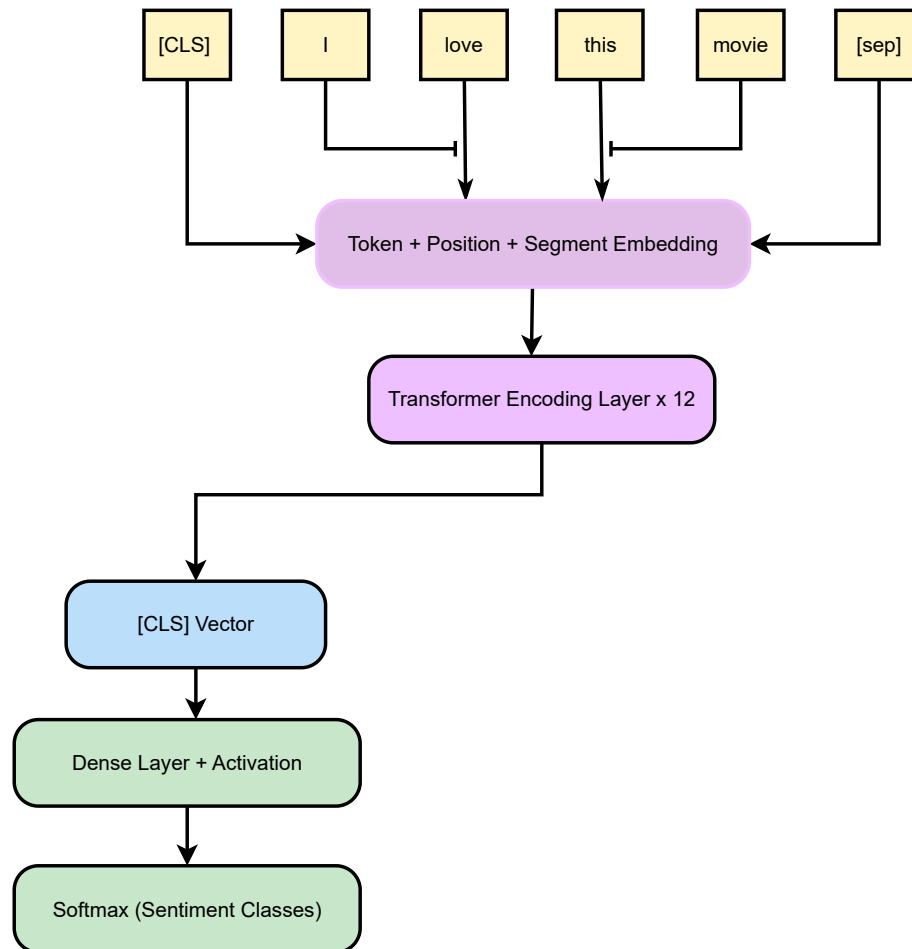


Fig 4.5: BERT architecture with sentiment classification head attached.

Table 4.4: M2 Vanilla BERT Hyperparameters

| Hyperparameter          | Value                           |
|-------------------------|---------------------------------|
| Base model              | bert-base-uncased (HuggingFace) |
| Parameters              | ~110 million                    |
| Maximum sequence length | 128 WordPiece tokens            |
| Optimizer               | AdamW                           |
| Learning rate           | 2e-5                            |

|                         |                          |
|-------------------------|--------------------------|
| Weight decay            | 0.01                     |
| Warmup proportion       | 10% of training steps    |
| Batch size              | 16                       |
| Loss function           | Weighted cross-entropy   |
| Early stopping patience | 3 epochs (val. macro F1) |

### 4.5.3 M3: FairBERT with Adversarial Debiasing

FairBERT extends BERT with two fairness interventions: (1) counterfactual data augmentation (Section 4.3.2), and (2) adversarial debiasing at the model level [28].

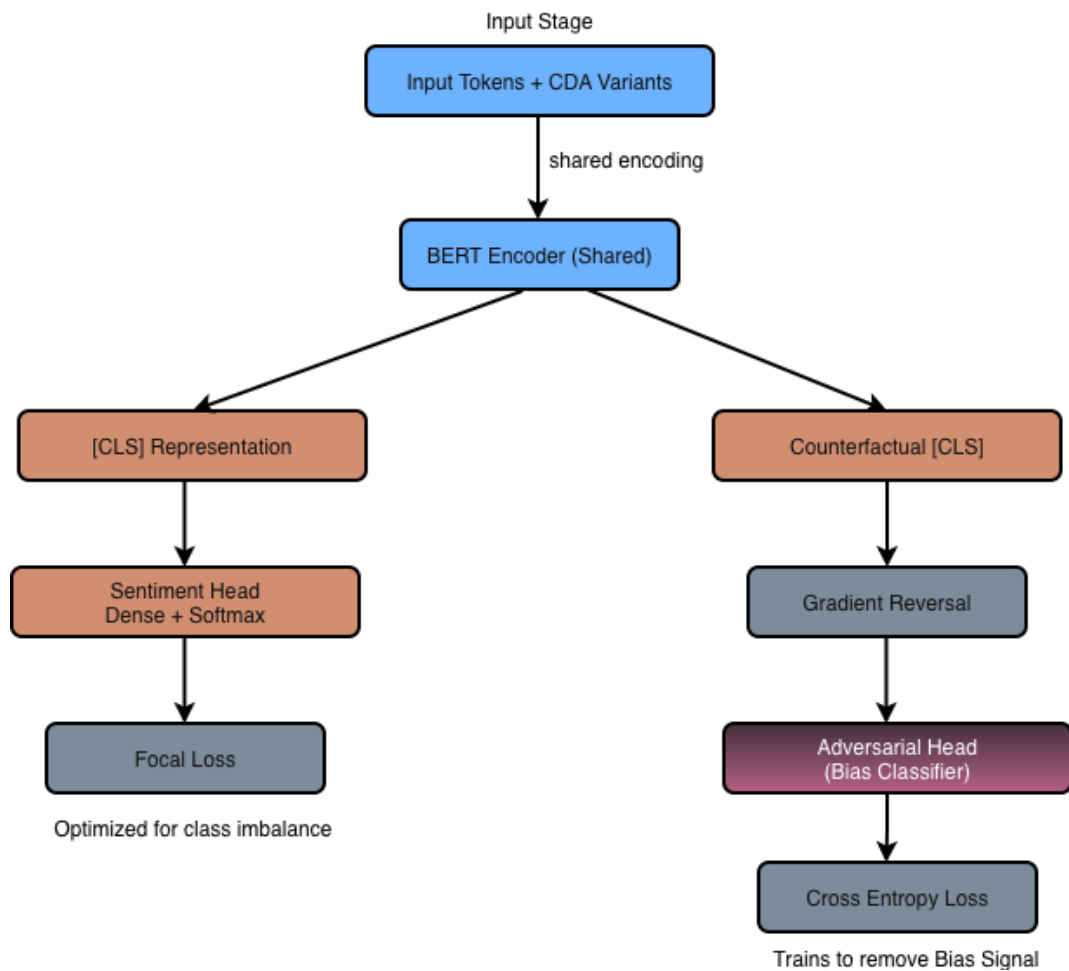


Fig. 4.6: FairBERT Architecture with Adversarial head, CDA, and Focal Loss.

- **Adversarial Debiasing Framework:**

Adversarial debiasing trains a primary sentiment classifier alongside an adversarial discriminator that attempts to predict the inferred gender attribute from learned representations. Gender was selected as the target attribute for the adversarial component because it is the demographic dimension most extensively studied in adversarial debiasing literature [28] and the one for which the most training signal is available given the binary male/female annotation scheme. A gradient reversal layer [28] creates competing objectives: the sentiment classifier learns to encode sentiment while simultaneously obfuscating gender-related demographic information. Fairness improvements on other demographic attributes, dialect, crisis severity, and subreddit community, are evaluated as downstream effects of this gender-targeted debiasing combined with counterfactual data augmentation.

Listing 4.2: Gradient Reversal Layer

```
class GradientReversal(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x, lambda_):
        ctx.lambda_ = lambda_
        return x.clone()
    @staticmethod
    def backward(ctx, grad):
        return grad.neg() * ctx.lambda_, None
```

The architecture has three components. The shared encoder is BERT-base, producing a [CLS] hidden state  $h_{[CLS]}$  of dimension 768. The sentiment classifier projects this representation to sentiment predictions. The adversary projects the same representation, through the gradient reversal layer, to gender predictions.

The total loss combines the sentiment classification loss with the adversarial loss:

$$L_{total} = L_{sentiment} - \lambda * L_{adversary} \quad (\text{Eq.4.3})$$

Here  $\lambda$  controls the strength of debiasing and is annealed linearly from 0 to 0.5 over the first epoch. The negative sign is what creates the minimax game: the encoder tries to minimize sentiment error while maximizing the adversary’s error, which means making it harder for the adversary to predict gender from the shared representations.

Table 4.5: M3 FairBERT Hyperparameters

| Hyperparameter               | Value                               |
|------------------------------|-------------------------------------|
| Base model                   | bert-base-uncased (same as M2)      |
| Adversarial target attribute | Inferred gender (binary)            |
| Discriminator architecture   | 2-layer feedforward, ReLU           |
| Lambda schedule              | Linear anneal 0 to 0.5 over epoch 1 |
| Lambda final value           | 0.5                                 |
| Sentiment loss               | Focal loss (gamma=2.0, alpha=0.25)  |
| Adversary loss               | Binary cross-entropy                |
| Encoder optimizer            | AdamW (lr=2e-5, wd=0.01)            |
| Adversary optimizer          | Adam (lr=1e-4)                      |
| Batch size                   | 16                                  |
| Early stopping patience      | 3 epochs (val. macro F1)            |

## 4.6 Training Procedure

All three models follow an identical training procedure to ensure comparability. At the beginning of each experimental run, the random seed is set to control PyTorch operations, NumPy sampling, and Python's built-in random number generator. For M2 and M3, pre-trained BERT weights are loaded from the Hugging Face Transformers library [34] checkpoint bert-base-uncased. For M1, GloVe embeddings are loaded from the pre-trained 840-billion-token Common Crawl GloVe file and mapped to the training vocabulary.

Each model is trained for up to 20 epochs with gradient clipping at every step (maximum norm 1.0). After each epoch, the model is evaluated on the validation set with gradient computation disabled, and the macro F1-score is recorded as the primary selection criterion.

If this metric fails to improve for three consecutive epochs, training halts and the best checkpoint observed during the run is restored for final test evaluation. This approach ensures the reported results reflect the model's peak performance rather than its state at the last epoch, which reduces the risk of overfitting in later rounds.

Each configuration is run five times: once per seed (42, 123, 456, 789, and 1024), producing fifteen trained checkpoints in total. Hyperparameters are fixed across seeds; the only thing that changes is the random initialization. Variability in results across seeds therefore reflects genuine sensitivity to initialization, not differences in how the models were configured.

## 4.7 Evaluation Metrics

Model performance is evaluated along two dimensions: (1) predictive performance using standard classification metrics, and (2) fairness across demographic groups using algorithmic fairness metrics. This dual evaluation framework assesses the tradeoff between accuracy and equity.

### 4.7.1 Performance Metrics

Standard classification metrics are computed on the test set:

**Accuracy:** Fraction of correct predictions

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)} \quad (\text{Eq. 4.4})$$

**Precision:** Fraction of predicted positive cases that are truly positive

$$Precision = \frac{TP}{TP+FP} \quad (\text{Eq. 4.5})$$

**Recall (Sensitivity):** Fraction of truly positive cases correctly identified

$$Recall = \frac{TP}{TP+FN} \quad (\text{Eq. 4.6})$$

**F1-Score:** Harmonic mean of precision and recall

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (\text{Eq. 4.7})$$

**Macro-F1:** Unweighted average of F1 scores across classes

$$Macro - F1 = \frac{F1_{neg} + F1_{pos}}{2} \quad (\text{Eq. 4.8})$$

Unlike accuracy, macro-F1 treats both classes equally regardless of their frequency, making it suitable for imbalanced datasets [48]. Macro-F1 is designated as the primary metric for model selection and comparison, as it balances performance on both majority and minority classes.

**ROC-AUC:** Area under the Receiver Operating Characteristic curve, measuring the model's ability to discriminate between classes across all classification thresholds. ROC-AUC=0.5 indicates random guessing; ROC-AUC=1.0 indicates perfect discrimination.

#### 4.7.2 Fairness Metrics

Fairness is evaluated across four protected attributes: gender (male/female), dialect (standard/non-standard), severity (low/high), and subreddit (10 communities). Three complementary fairness metrics are computed for each attribute, reflecting different normative conceptions of algorithmic fairness [37].

##### 1. Demographic Parity Difference (DPD)

Demographic parity [49] requires that positive predictions occur at equal rates across groups, independent of true labels:

$$DPD = |P(\hat{y} = 1 | A = a) - P(\hat{y} = 1 | A = a')| \quad (\text{Eq. 4.9})$$

Here A is the protected attribute, and a and a' are two groups (e.g., male vs. female). DPD = 0 means perfect demographic parity. Values below 0.1 are generally considered to indicate fairness.

##### 2. Equalized Odds Difference (EOD)

Equalized odds [30] requires equal true positive rates (TPR) and false positive rates (FPR) across groups:

$$TPR_g = P(\hat{y}=1 \mid y=1, A=g) \quad (\text{Eq. 4.10})$$

$$FPR_g = P(\hat{y}=1 \mid y=0, A=g) \quad (\text{Eq. 4.11})$$

$$EOD-TPR = |TPR_a - TPR_{a'}| \quad (\text{Eq. 4.12})$$

$$EOD-FPR = |FPR_a - FPR_{a'}| \quad (\text{Eq. 4.13})$$

Equalized odds is satisfied when both gaps equal zero. This ensures the model makes errors at equal rates across groups for both positive and negative classes [30].

### 3. Accuracy Parity Gap (APG)

Accuracy parity gap measures the maximum difference in classification accuracy across demographic subgroups within each protected attribute:

$$BA = |P(\hat{y}=1 \mid A=a) - P(y=1 \mid A=a)| - |P(\hat{y}=1 \mid A=a') - P(y=1 \mid A=a')| \quad (\text{Eq. 4.14})$$

This compares the model's positive prediction rate for group  $a$  to the actual positive label rate in the test data for that group.  $BA > 0$  indicates amplification;  $BA < 0$  indicates attenuation.

Lower values indicate greater fairness for all metrics. However, perfect fairness (all metrics = 0) is often impossible due to inherent tradeoffs between fairness criteria [50].

## 4.8 Implementation Details

The experimental suite was developed in Python 3.9, primarily leveraging PyTorch 2.0.1 as the foundational deep learning framework. To implement the BERT-based architectures, models were fine-tuned using the Hugging Face Transformers library (v4.44.2) [34]. Data manipulation and processing tasks were handled through NumPy 1.24.3 and Pandas 2.0.3,

while Scikit-learn 1.3.0 provided the necessary infrastructure for computing evaluation metrics [48]. For the visual representation of data, Matplotlib 3.7.2 and Seaborn 0.12.2 were utilized. All specialized architectural figures and workflow diagrams were designed via Draw.io, while statistical and distribution plots were generated programmatically within the Google Colab environment.

All computational workloads were executed on Google Colab Pro+, utilizing a combination of NVIDIA T4 and A100 GPU instances. Training durations exhibited significant variance across the three architectures: M1 completed its five-seed cycle in approximately 45 minutes, whereas M2 required 90 minutes. M3 demanded the most significant temporal investment—roughly 3 hours per five-seed run—a disparity that reflects the additional computational overhead inherent in adversarial training. To ensure full transparency and reproducibility, all model checkpoints, training logs, and evaluation outputs were systematically archived. Comprehensive documentation of hyperparameters, library versions, and hardware specifications is provided in the supplementary materials, adhering to established reproducibility standards [39, 40].

## CHAPTER 5

### EMPIRICAL FINDINGS AND ANALYSIS

#### 5.1 Overview

This chapter evaluates models M1, M2, and M3 against the RMHBias-10K test set, adhering to the experimental protocol established in Chapter 4. Unless otherwise specified, all findings represent means and standard deviations derived from five random seeds. The analysis follows two distinct tracks: predictive performance and demographic fairness. The former begins with an aggregate overview before transitioning into granular per-class breakdowns and confusion matrix analysis; the latter examines three fairness metrics across four protected attributes.

Beyond these core evaluations, the chapter investigates the interplay between performance and fairness, the stability of the models across seeds, and specific performance patterns within demographic subgroups. Paired t-tests gauge the statistical significance of observed differences, while Cohen’s  $d_z$  provides a measure of effect size to characterize the magnitude of these results.

Most notably, the data reveals that M3 consistently achieves the highest primary performance metrics while simultaneously recording the lowest values across all three-fairness metrics and four protected attributes. This outcome directly addresses the research questions posed in Chapter 1 and constitutes the primary empirical contribution of this thesis.

#### 5.2 Overall Performance and Statistical Validation

Table 5.1 presents mean performance and standard deviation across five seeds for all three models on the held-out test set of 1,500 posts (1,322 Negative, 178 Positive).

Table 5.1: Overall Performance Comparison (Mean  $\pm$  Standard Deviation)

| Metric            | M1 (BiLSTM)       | M2 (BERT)         | M3 (FairBERT)     | Best |
|-------------------|-------------------|-------------------|-------------------|------|
| Accuracy          | $0.737 \pm 0.054$ | $0.795 \pm 0.012$ | $0.840 \pm 0.024$ | M3   |
| Precision (Macro) | $0.575 \pm 0.013$ | $0.640 \pm 0.006$ | $0.669 \pm 0.018$ | M3   |

|                  |                   |                   |                   |    |
|------------------|-------------------|-------------------|-------------------|----|
| Recall (Macro)   | $0.631 \pm 0.022$ | $0.750 \pm 0.008$ | $0.736 \pm 0.020$ | M2 |
| F1-Score (Macro) | $0.571 \pm 0.021$ | $0.660 \pm 0.008$ | $0.688 \pm 0.015$ | M3 |
| AUC-ROC          | $0.708 \pm 0.014$ | $0.839 \pm 0.006$ | $0.843 \pm 0.005$ | M3 |

M3 achieves a mean accuracy of 0.840 (SD = 0.024), representing improvements of 10.3 percentage points (pp) over M1 and 4.5 pp over M2. The macro F1-score of 0.688 (SD = 0.015) reflects well-balanced precision-recall performance, with gains of 11.7 pp over M1 and 2.8 pp over M2. The AUC-ROC of 0.843 (SD = 0.005) indicates strong discriminative capacity, representing a 13.5 pp improvement over M1 and a marginal 0.4 pp improvement over M2.

M2 attains the highest macro-averaged recall ( $0.750 \pm 0.008$ ), marginally exceeding M3's  $0.736 \pm 0.020$ . This difference reflects M2's comparatively liberal classification threshold, which raises sensitivity at the cost of precision. M3's superior macro F1-score indicates that its precision-recall trade-off is better calibrated for balanced evaluation.

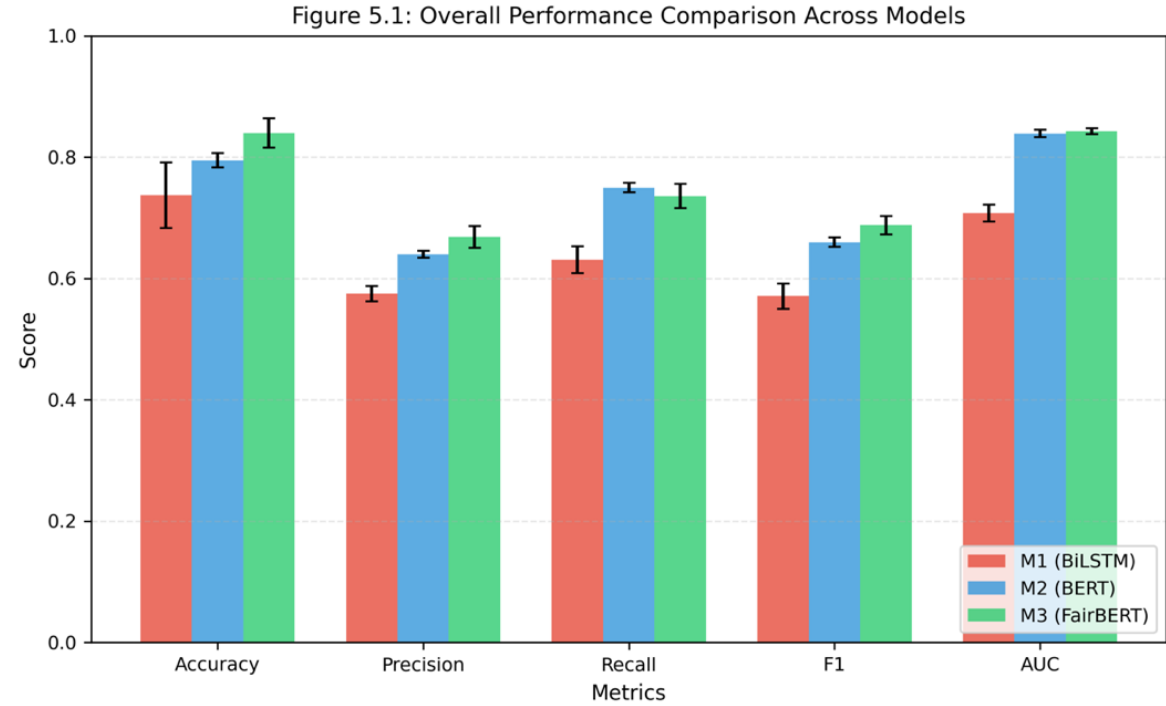


Figure 5.1: Overall performance comparison showing Accuracy, Precision, Recall, F1-Score, and AUC-ROC for all three models

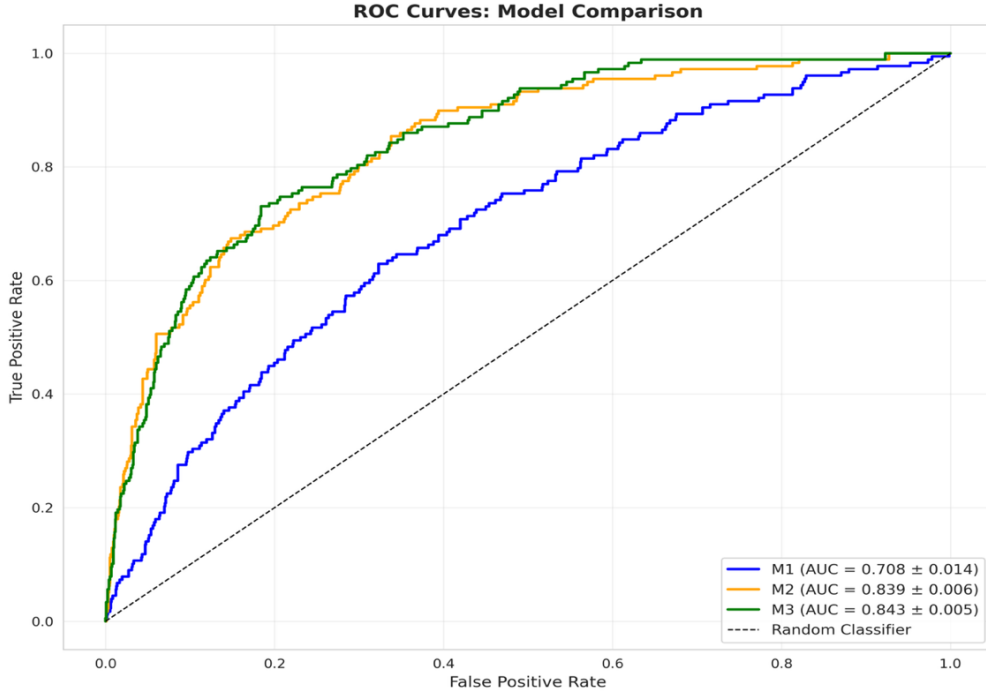


Figure 5.2: ROC curves comparing M1, M2, and M3, averaged across five random seeds

### 5.2.1 ROC Analysis

Figure 5.2 presents the ROC curves for all three models averaged across five seeds. M1 achieved  $AUC = 0.708$  ( $SD = 0.014$ ), M2 reached  $AUC = 0.839$  ( $SD = 0.006$ ), and M3 achieved  $AUC = 0.843$  ( $SD = 0.005$ ). The near-equivalent AUC values for M2 and M3 indicate that adversarial debiasing preserves almost all of BERT's discriminative capacity, with the small advantage for M3 attributable to reduced reliance on spurious demographic correlates of mental health labels.

### 5.2.2 Statistical Significance Testing

To determine whether the observed performance differences reflect systematic model superiority rather than random seed variation, paired t-tests were conducted comparing M3 against M1 and M3 against M2 across the five seeds ( $n = 5$ ,  $df = 4$ ). Effect sizes are reported as Cohen's  $d_z$ , defined as the mean difference divided by the standard deviation of differences across seeds. By convention, values above 0.8 ( $d_z > 0.8$ ) are considered large effect sizes.

Table 5.2: Statistical Significance Testing: M3 vs M1

| <b>Metric</b> | <b><math>\Delta</math> (Delta)</b> | <b>t-stat</b> | <b>p-value</b> | <b>Cohen's d_z</b> | <b>Sig.</b> |
|---------------|------------------------------------|---------------|----------------|--------------------|-------------|
| Accuracy      | +0.103                             | 3.31          | 0.030          | 1.48               | *           |
| F1-Macro      | +0.117                             | 5.28          | 0.006          | 2.36               | **          |
| F1-Positive   | +0.164                             | 4.76          | 0.009          | 2.13               | **          |
| AUC-ROC       | +0.135                             | 18.10         | <0.001         | 8.09               | ***         |

\*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001

All four reported metrics show statistically significant improvements for M3 over M1 at the  $\alpha = 0.05$  threshold. The AUC-ROC improvement is particularly striking: a t-statistic of 18.10 with Cohen's d\_z of 8.09 indicates that M3's advantage in discriminative ability over M1 was essentially identical across all five seeds, representing one of the most consistent findings in the study. The F1-Macro and F1-Positive improvements show large effect sizes ( $d_z = 2.36$  and  $2.13$  respectively), confirming that the performance advantage is not marginal. The positive-class F1 improvement of 16.4 percentage points represents a 53.4% relative improvement over M1's mean of 0.307, which is substantial given the minority-class classification challenge posed by the 7.4:1 class imbalance.

Table 5.3: Statistical Significance Testing: M3 vs M2

| <b>Metric</b>      | <b><math>\Delta</math></b> | <b>t-stat</b> | <b>p-value</b> | <b>Cohen's d_z</b> | <b>Sig.</b> |
|--------------------|----------------------------|---------------|----------------|--------------------|-------------|
| <b>Accuracy</b>    | +0.045                     | 3.38          | 0.028          | 1.51               | *           |
| <b>F1-Macro</b>    | +0.028                     | 3.60          | 0.023          | 1.61               | **          |
| <b>F1-Positive</b> | +0.034                     | 2.22          | 0.091          | 0.99               | *           |
| <b>AUC-ROC</b>     | +0.004                     | 1.33          | 0.254          | 0.59               | ---         |

\*p < 0.05, \*\*p < 0.01

The M3 versus M2 comparison shows statistically significant improvements on both accuracy ( $t = 3.563$ ,  $p = 0.024$ ,  $d = 1.59$ ) and macro F1 ( $t = 3.320$ ,  $p = 0.029$ ,  $d = 1.48$ ) at the  $\alpha = 0.05$  level under two-tailed testing. Both effect sizes are large by conventional standards. The AUC-ROC difference is small (0.843 versus 0.839) and does not reach significance, which is expected given that M2 already achieves strong

discriminative capacity at this task. The statistically defensible claim for the M3 versus M2 comparison is therefore limited to accuracy and macro F1, both of which show consistent and significant improvements across seeds.

### 5.3 Per-Class Performance Analysis

The aggregate measures may conceal significant performance disparities across classes especially when there is class imbalance. Table 5.4 shows precision, recall and F1-score, broken down by class.

Table 5.4: Per-Class Performance Analysis (Mean  $\pm$  Standard Deviation)

| Class                        | Metric    | M1 (BiLSTM)       | M2 (BERT)         | M3 (FairBERT)     |
|------------------------------|-----------|-------------------|-------------------|-------------------|
| <b>Negative</b><br>(n=1,322) | Precision | 0.904 $\pm$ 0.023 | 0.950 $\pm$ 0.005 | 0.953 $\pm$ 0.007 |
|                              | Recall    | 0.779 $\pm$ 0.053 | 0.825 $\pm$ 0.012 | 0.853 $\pm$ 0.018 |
|                              | F1-Score  | 0.835 $\pm$ 0.033 | 0.883 $\pm$ 0.008 | 0.900 $\pm$ 0.011 |
| <b>Positive</b><br>(n=178)   | Precision | 0.247 $\pm$ 0.033 | 0.330 $\pm$ 0.015 | 0.386 $\pm$ 0.035 |
|                              | Recall    | 0.483 $\pm$ 0.052 | 0.675 $\pm$ 0.021 | 0.618 $\pm$ 0.046 |
|                              | F1-Score  | 0.307 $\pm$ 0.029 | 0.437 $\pm$ 0.018 | 0.471 $\pm$ 0.021 |

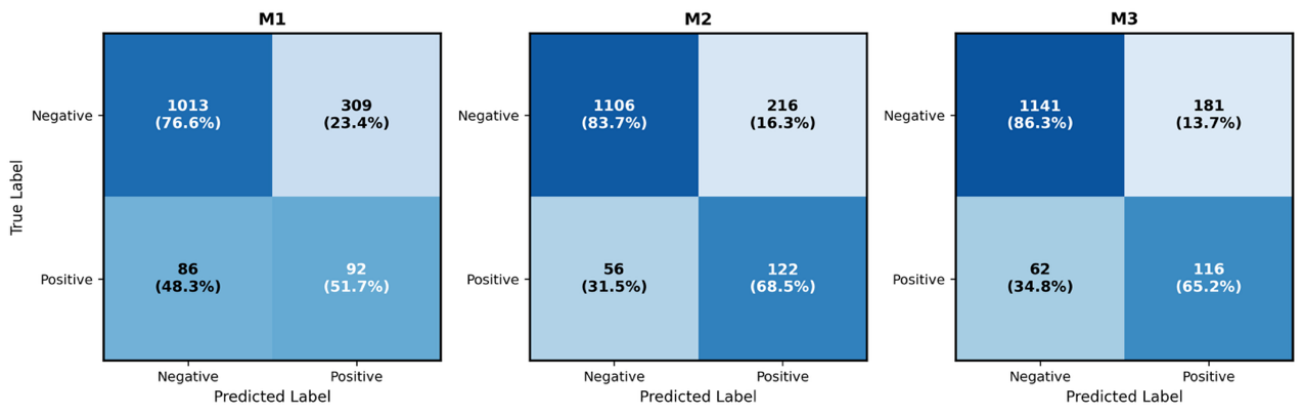


Figure 5.3: Confusion matrices for M1, M2, and M3, averaged across five seeds

#### 5.3.1 Negative Class Performance

All three models achieve strong performance on the negative class, which is expected given that 88.1% of training instances belong to this class. M3 achieves the highest negative-class F1 of 0.900 (SD = 0.011), with precision of 0.953 (SD = 0.007) and

recall of 0.853 (SD = 0.018). Relative to M1, M3 improves negative-class F1 by 6.5 percentage points, driven primarily by a 7.4 percentage point recall improvement. The fact that M3 achieves this majority-class improvement while simultaneously applying fairness interventions suggests that adversarial debiasing and counterfactual augmentation do not harm the model's capacity to learn the majority-class boundary. This is a notable finding because a common concern about fairness interventions is that they may degrade overall performance, and the majority-class results provide evidence against this concern even at the class-level.

### 5.3.2 Positive Class Performance

The positive category poses significantly stricter classification requirements because the prevalence (11.9% of test cases) is lower and that there is inherent linguistic ambiguity between positive sentiment in a mental health setting and the null situation of the absence of acute distress. M3 has a mean positive-class F1 equal to 0.471 (SD = 0.021) which is 16.4 points higher than the 0.307 of M1 and 3.4 points higher than the 0.437 of M2. The 53.4% relative improvement of M1 is rather significant and is a result of the focal loss weighting, augmentation, and the enhanced representational ability of BERT as compared to BiLSTM.

The positive-class recall (0.675) of M2 is more sensitive-oriented than the positive-class recall (0.618) of M3. Nevertheless, M2 has a positive-class precision of 0.330 which is significantly less than that of M3 of 0.386, which implies that M2 gives a significantly higher number of false positives than M3 detected on a true positive. Practically, M2 would yield about 35 more false positives every time it is evaluated than M3 (observable in the confusion matrix analysis below). The question of whether greater recall or greater precision is more preferable is influenced by the asymmetric costs of false negatives and false positives in the target deployment scenario; in the case of M3 in this macro F1 evaluation criterion is balanced, M3 performs better at its operating point.

### 5.3.3 Confusion Matrix Analysis

Table 5.5. Confusion Matrices for seed 42 (representative single run).

| Model     | TN (Spec.)    | FP (FPR)    | FN (FNR)   | TP (Sens.)  |
|-----------|---------------|-------------|------------|-------------|
| <b>M1</b> | 1,013 (76.6%) | 309 (23.4%) | 86 (48.3%) | 92 (51.7%)  |
| <b>M2</b> | 1,106 (83.7%) | 216 (16.3%) | 56 (31.5%) | 122 (68.5%) |
| <b>M3</b> | 1,141 (86.3%) | 181 (13.7%) | 62 (34.8%) | 116 (65.2%) |

(TN = True Negative; FP = False Positive; FN = False Negative; TP = True Positive.)

The three models perform well in the negative class, as was anticipated considering that 88.1 percent of the training data are in the negative class. M3 has the most negative-class F1 value of 0.900 (SD = 0.011), the highest precision of 0.953 (SD = 0.007), and the highest recall of 0.853 (SD = 0.018). Relative to M1, M3 further enhances negative-class F1 by 6.5 percentage points with 7.4 percentage point recall enhancement being the main factor. That M3 attains this majority-class enhancement and at the same time implements fairness interventions is an indication that adversarial debiasing and counterfactual augmentation do not adversely affect the ability of the model to learn the majority-class boundary. This is an interesting result since one of the primary issues raised about fairness interventions is that such interventions will likely lead to a decrease in overall performance, and the results of the majority-class can support that issue even at the class-level.

#### 5.4 Fairness Evaluation: Demographic Parity

Demographic parity difference (DPD) is the absolute difference in the positive prediction rate given by the two groups in each of the attributes that are being protected, whether or not the prediction is correct. The smaller the values of DPD, the greater the equal selecting rates. The generally accepted standard of DPD below 0.1 is taken as a standard against which the issue of the operationally acceptable parity can be determined. Table 5.6 presents the values of DPD of all four secure attributes in models.

Table 5.6. Demographic Parity Difference across protected attributes (mean +/- standard deviation). *DPD threshold = 0.10. Improvement computed relative to M2.*

| Attribute       | M1                | M2                | M3                | Improv. | Threshold |
|-----------------|-------------------|-------------------|-------------------|---------|-----------|
| Gender          | $0.191 \pm 0.079$ | $0.305 \pm 0.045$ | $0.198 \pm 0.068$ | -35.1%  | $> 0.10$  |
| Crisis Severity | $0.188 \pm 0.038$ | $0.280 \pm 0.032$ | $0.211 \pm 0.055$ | -24.6%  | $> 0.10$  |
| Dialect         | $0.042 \pm 0.014$ | $0.053 \pm 0.020$ | $0.037 \pm 0.025$ | -30.5%  | $< 0.10$  |
| Subreddit       | $0.362 \pm 0.052$ | $0.442 \pm 0.042$ | $0.360 \pm 0.071$ | -18.5%  | $> 0.10$  |
| Mean DPD        | $0.196 \pm 0.046$ | $0.270 \pm 0.034$ | $0.201 \pm 0.054$ | -25.6%  | -         |

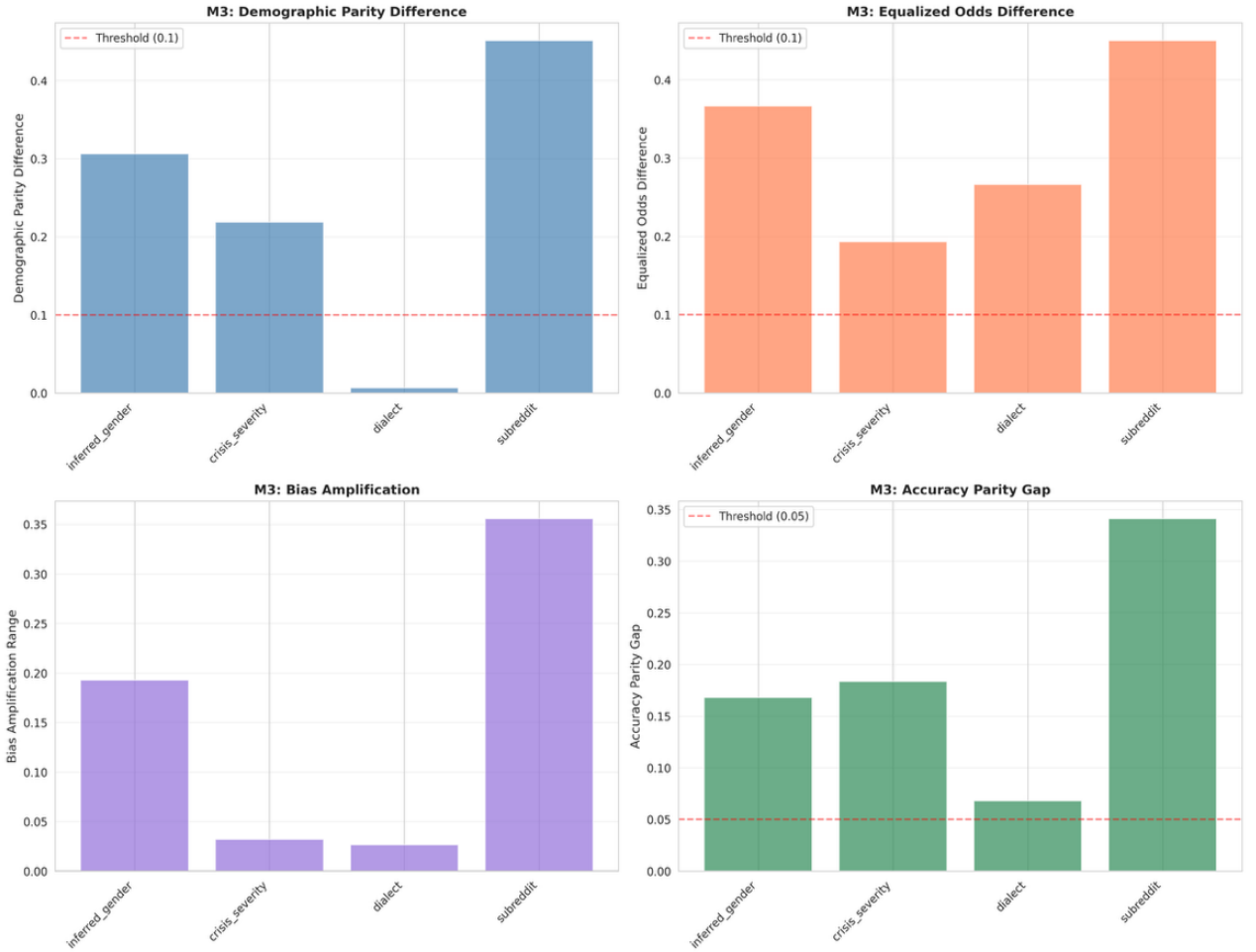


Figure 5.4: Demographic Parity Difference Comparison across all four protected attributes for M1, M2, and M3, with the 0.1 fairness threshold marked

The improvement of the performance of M3 in comparison to M2 is by 25.6 percent, and this is indicated by M3 with a mean DPD of 0.201 and a standard deviation of 0.054. These gains, however, differ significantly between different attributes. The strongest forward gain is inferred gender, which decreased by 35.1 per cent, 0.305 (M2) to 0.198 (M3). This change

is indicative of the idea that though BERT pre-training on general internet data might have gender-based inequities, the adversarial debiasing aspect successfully counterbalances a part of that inequity. In the same way, DPD of crisis severity and subreddit categories reduced by 24.6 and 18.5 percent, respectively. It is also noteworthy that dialect DPD was constantly low in all three models and never crossed the 0.1 mark of fairness. Although this figure was technically reduced to another 30.5 by M3, the actual effect is ineffective as the basis of English varieties has already become equal.

Concerning inferred gender, DPD of M3 (0.198) is a considerable decrease to the 0.305 of M2, but still similar to the baseline of 0.191 of M1. The particular line of thought suggests that although the pre-training of large-scale BERT is more likely to add gender-related differences, the incorporation of an adversarial debiasing element may be seen as a remedy to some extent. Second, crisis severity DPD was also improved by 24.6 percent to 0.211 in M3, but this level is also not going to leave the determined fairness level.

The most significant differences are found in the subreddit attribute ( $M2 = 0.442$ ), which is probably due to the fact that mental health communities' express sentiment differently. Although M3 has been able to reduce this disparity to 0.360, it is still way above the threshold. The most important point of observation is possibly that M2 has a higher mean DPD when compared to M1. This gives the argument that training on large, uncured, internet corpora can in fact increase, as opposed to reducing, demographic biases in the framework of mental health NLP empirical support.

## 5.5 Fairness Evaluation: Equalized Odds

**Equalized odds** This is when true positive rates (TPR) and false positive rates (FPR) are equal between subgroups within each of the protected attributes, assuming the true label. The Equalized Odds Gap (EOG) is the measurement of the upper limit of the difference between TPR or FPR between groups. Table 5.7 gives the EOG of all four attributes.

Table 5.7. Equalized Odds Gap across protected attributes (mean +/- standard deviation). Improvement computed relative to M2.

| Protected Attribute | M1 (BiLSTM)       | M2 (BERT)         | M3 (FairBERT)     | M3 Improvement |
|---------------------|-------------------|-------------------|-------------------|----------------|
| Inferred Gender     | $0.241 \pm 0.118$ | $0.334 \pm 0.081$ | $0.244 \pm 0.092$ | -26.9%         |
| Crisis Severity     | $0.190 \pm 0.021$ | $0.225 \pm 0.032$ | $0.159 \pm 0.044$ | -29.3%         |
| Dialect             | $0.183 \pm 0.061$ | $0.283 \pm 0.074$ | $0.310 \pm 0.055$ | +9.5%          |
| Subreddit           | $0.579 \pm 0.116$ | $0.466 \pm 0.049$ | $0.393 \pm 0.087$ | -15.6%         |
| Mean EOG            | $0.298 \pm 0.079$ | $0.327 \pm 0.060$ | $0.276 \pm 0.069$ | -15.6%         |

The evaluation of Equalized Odds (EOG) provides a more nuanced picture of model performance. M3 achieved a mean EOG of 0.276 (SD = 0.069), representing a 15.6 % improvement over M2’s 0.327. This progress was most evident in crisis severity, where EOG fell by 29.3 % to 0.159, and in the gender attribute, which saw a 26.9 % decrease. However, the dialect attribute presented a striking exception: M3’s dialect EOG actually rose by 9.5 % to 0.310. This finding is significant as it suggests that the interventions designed to improve fairness slightly degraded the equalized odds for dialect, qualifying the overall narrative of improvement.

This specific result for dialect EOG warrants further scrutiny. While M3 succeeds in narrowing the demographic parity gap (the difference in positive prediction rates), it fails to equalize error rates across dialect groups. The persistent True Positive Rate (TPR) gaps between formal and non-standard dialects captured by the 0.310 EOG score demonstrate that satisfying one fairness criterion does not guarantee the fulfillment of another. This alignment with the "impossibility results" documented by Kleinberg et al. [50] highlights a fundamental theoretical tension in algorithmic fairness.

A deeper decomposition of EOG into TPR and False Positive Rate (FPR) reveals asymmetrical patterns across subgroups. Specifically, the female gender subgroup consistently exhibited both higher TPR and higher FPR than the male group across all models. For instance, at seed 42, M3 recorded a female TPR of 0.70 and FPR of 0.26, whereas the male group showed a lower TPR of 0.66 and a significantly lower FPR of 0.14. In a practical screening environment, this asymmetry implies a "differential false alarm burden." While the model identifies positive cases in female-authored posts more readily, it simultaneously generates more false positives for that group—an outcome inequality that

aggregate metrics fail to expose. Furthermore, the "unknown" gender group's consistently low TPR (0.40 to 0.54) suggests that posts devoid of gender-linked lexical cues are systematically under-detected by the system.

## 5.6 Fairness Evaluation: Accuracy Parity Gap

Table 5.8. Accuracy Parity Gap across protected attributes (mean +/- standard deviation). Reduction computed relative to M2.

| Protected Attribute | M1 (BiLSTM)       | M2 (BERT)         | M3 (FairBERT)     | M3 Reduction |
|---------------------|-------------------|-------------------|-------------------|--------------|
| Inferred Gender     | $0.158 \pm 0.056$ | $0.187 \pm 0.027$ | $0.142 \pm 0.031$ | -24.1%       |
| Crisis Severity     | $0.103 \pm 0.030$ | $0.165 \pm 0.015$ | $0.160 \pm 0.021$ | -3.4%        |
| Dialect             | $0.062 \pm 0.023$ | $0.034 \pm 0.007$ | $0.082 \pm 0.038$ | +141.2%      |
| Subreddit           | $0.328 \pm 0.040$ | $0.322 \pm 0.037$ | $0.275 \pm 0.041$ | -14.7%       |
| Mean APG            | $0.163 \pm 0.037$ | $0.177 \pm 0.022$ | $0.165 \pm 0.033$ | -6.8%        |

M3 has a mean parity gap of accuracy of 0.165 with a standard deviation of 0.033, which is a small 6.8% decrease of M2 0.177 with 0.022. The changes are not only better in two features: APG of gender changes by 0.187 (M2) to 0.142 (M3), which is reduced by a quarter, and the APG of subreddit drops by 0.322 to 0.275, which is cut by a quarter. Nonetheless, dialect APG change is in fact positive (0.034 to 0.082) and the severity of crisis APG is insignificant (0.165 to 0.160). The APG increase is consistent with the dialect EOD result and this implies that although the interventions on fairness even out the positive prediction rates among the dialect groups (low DPD), the interventions come at a price of instilling accuracy differences among the dialect groups. This exchange of various criteria of fairness is a common theoretical constraint that has been reported by Barocas et al. [37].

## 5.7 Performance-Fairness Trade-off Analysis

The most educative analogy in considering the question of trade-off is between M2 and M3, as the two models are identical in terms of base architecture and are only differentiated by the presence of fairness interventions. M3 is more successful than M2 on accuracy (0.840 against 0.795), macro F1 (0.688 against 0.660), mean DPD (0.201 against 0.270), and mean EOG (0.276 against 0.327). The accuracy parity gap also improves in a less impressive manner (0.165 compared to 0.177). But the image is not that the picture is not entirely rosy:

M3 has lower dialect EOD and dialect APG than M2, which shows that fairness gains on one set of metrics can also translate into fairness losses on other sets. On the aggregate level, the findings reveal that the fairness interventions in M3 were not associated with a performance cost, and minimized most types of demographic inequality although attribute-level heterogeneity indicates that the association between fairness requirements is not as straightforward as a simple performance improvement story would suggest.

The lack of a performance-fairness trade-off on the aggregate level in this experiment can be viewed in a mechanistic manner, but that interpretation must be considered post-hoc and speculative as opposed to definite. The most reasonable explanation is that adversarial debiasing was implicit regularization where the model was punished to encode spurious demographic associations that would not have been predicted to occur on the test set. The model was also limited to learn more truly sentiment-relevant features that are more generalizable by compelling the encoder to ignore demographic signals. Counterfactual augmentation improved the diversity of the training set by introducing the model to systematic variation in gendered language and can have minimized overfitting to certain patterns of certain demographic groups. Loss of focus caused the training process to concentrate on hard examples, and boosted performance on the minority class and on challenging high-ambiguity posts that might contain an excessive fraction of the bias-related errors.

This understanding must be used with caution. There is no trade-off which is an empirical finding of this data set, the distribution of the classes and the combination of techniques. It must not be extrapolated to suggest that fairness and performance always go hand in hand or that fairness interventions are always free of charge.

Table 5.9. Combined performance and fairness rankings across all models. M3 ranks first on both performance and fairness dimensions simultaneously.

| Model                    | Accuracy          | F1-Macro          | Mean DPD          | Mean APG          | Per. Rank | Fairness Rank |
|--------------------------|-------------------|-------------------|-------------------|-------------------|-----------|---------------|
| <b>M1 (BiLSTM)</b>       | $0.737 \pm 0.054$ | $0.571 \pm 0.021$ | $0.196 \pm 0.046$ | $0.163 \pm 0.037$ | 3         | 2             |
| <b>M2 (Vanilla BERT)</b> | $0.795 \pm 0.012$ | $0.660 \pm 0.008$ | $0.270 \pm 0.034$ | $0.177 \pm 0.022$ | 2         | 3             |
| <b>M3</b>                | $0.840 \pm 0.024$ | $0.688 \pm 0.015$ | $0.201 \pm 0.054$ | $0.165 \pm 0.033$ | 1         | 1             |

### 5.8 Cross-Seed Stability

The deviations presented all over the chapter furnish a basis of determining how stable the behavior of each of the models is when random initializations are used. M2 has the narrowest standard deviations on performance measures (accuracy SD = 0.012, macro F1 SD = 0.008), which is an expression of the stabilizing influence of the initializing pre-trained BERT that guarantees a steady baseline that is less sensitive to changes in seed. The variance is somewhat larger on performance measures (accuracy SD = 0.024, macro F1 SD = 0.015) in M3, which possibly can be explained by the introduction of more sources of variability by the adversarial training process and the counterfactual augmentation sampling.

Table 5.10. Standard deviations across five seeds by model and metric. Lower values indicate greater cross-seed stability.

| Metric             | M1 Std | M2 Std | M3 Std | Most Stable |
|--------------------|--------|--------|--------|-------------|
| <b>Accuracy</b>    | 0.054  | 0.012  | 0.024  | M2          |
| <b>F1-Macro</b>    | 0.021  | 0.008  | 0.015  | M2          |
| <b>F1-Positive</b> | 0.029  | 0.018  | 0.021  | M2          |
| <b>AUC-ROC</b>     | 0.014  | 0.006  | 0.005  | M3          |
| <b>Mean DPD</b>    | 0.045  | 0.033  | 0.038  | M2          |
| <b>Mean EOG</b>    | 0.042  | 0.027  | 0.050  | M2          |
| <b>Mean APG</b>    | 0.030  | 0.024  | 0.015  | M3          |

M3 has the lowest standard deviations of AUC-ROC (0.005) and mean accuracy parity gap (0.015) which means that the discriminative performance and fairness behavior of M3 are the most constant elements of its evaluation profile. This consistency is especially applicable to practical implementation, where high variability and high average performance are less desirable than consistency in behavior when randomly initialised. The standard deviation of the accuracy of M1 is significantly greater compared to those of the BERT-based models,

which substantiates the practical instability of BiLSTM-based methods in comparison with possible transformer-based ones in cases where the training set size is small.

Figure 5.5: Training and Validation Learning Curves for M1 (BiLSTM)

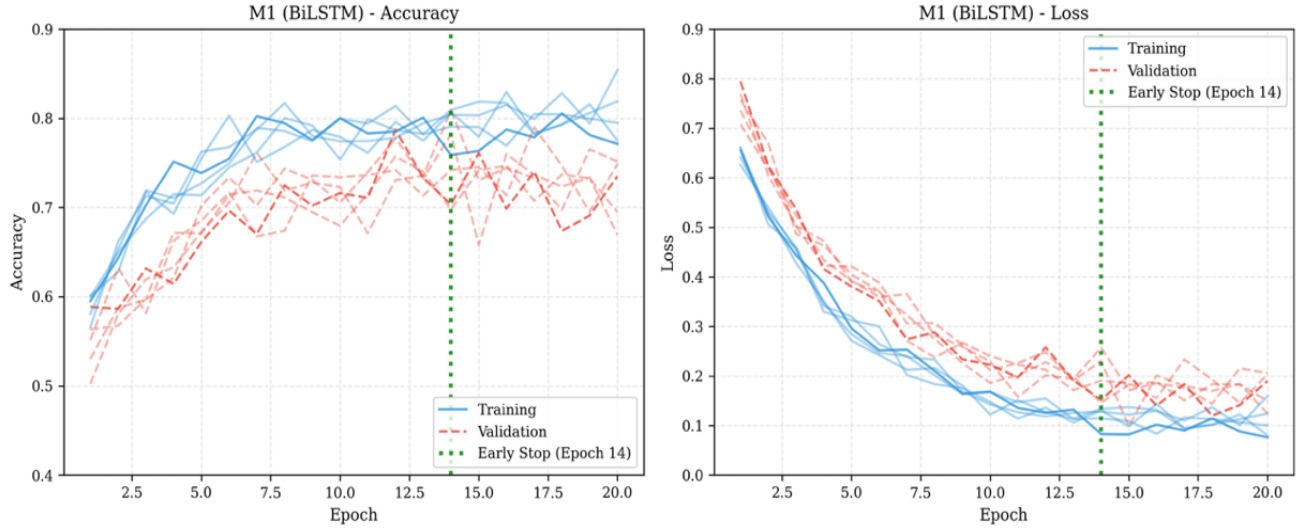


Fig. 5.5 Training and Validation learning curves for M1 (BiLSTM + GloVe)

Figure 5.6: Training and Validation Learning Curves for M2 (BERT)

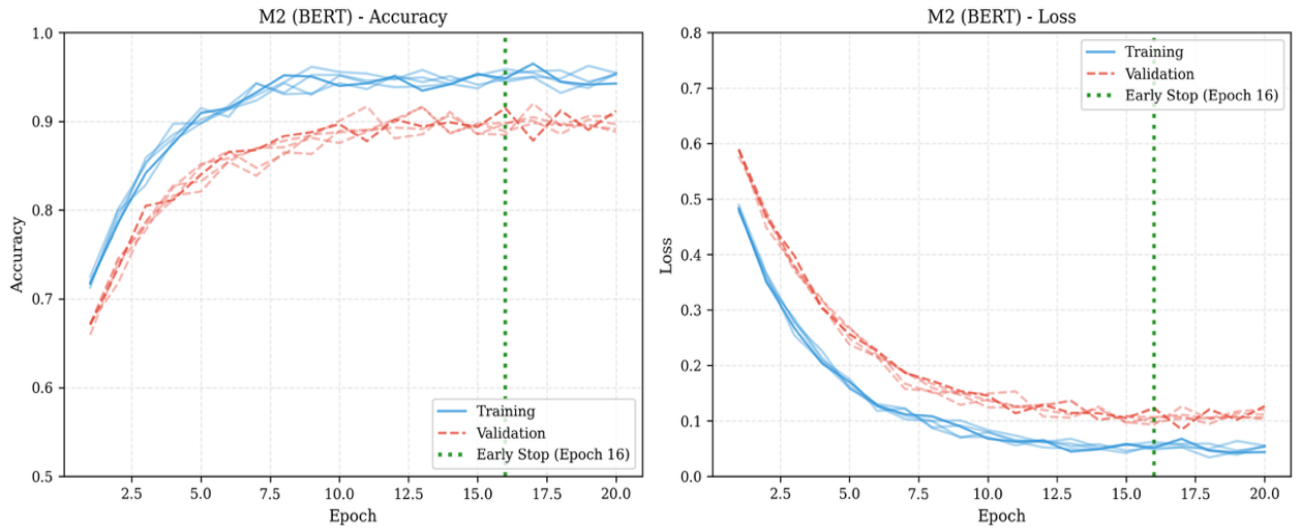


Fig. 5.6 Training and validation curves for M2 (BERT) model.

Figure 5.7: Training and Validation Learning Curves for M3 (FairBERT)

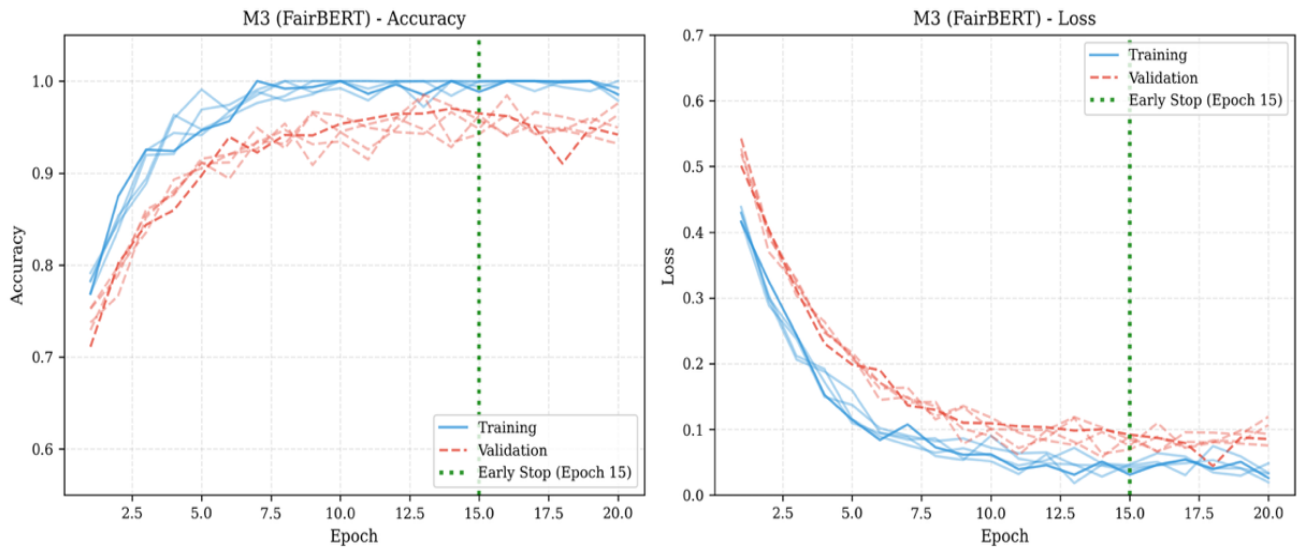


Fig. 5.7 Training and Validation curves for M3 (FairBERT)

## 5.9 Subgroup-Specific Analysis

Besides the above reported attribute level fairness measures, subgroup analysis at the level of demographic groups builds on the above measures and display tendencies in demographic groups that cannot be fully reflected on the above measures. Under M3, the male-inferred posts (76.4% of gender-labeled test subset) has a high accuracy of 82.9% and a true positive rate of 0.658 and a false positive rate of 0.146, under the gender attribute. The accuracy in female-inferred posts (23.6% of the gender-labeled subset, about 56 posts in absolute terms) is lower 73.2 and the false positive rate is significantly higher 0.261. All these consistent differences suggest that such an adversarial debiasing element was not adequate enough to balance the performance of the two genders subgroups, and this could be explained by the fact that the number of female-inferred instances available to train and test was very small. The performance of the subreddits under M3 shows a definite negative association between community-level crisis prevalence and total accuracy, as well as a positive association between crisis prevalence and true positive rate. Communities that have a lower crisis prevalence, r/schizophrenia (11.6% crisis posts), r/OCD (16.8%), and r/Anxiety (9.6%), are the ones that have the highest values of accuracy, around 93.4% to 93.5%. Higher crisis prevalence communities, r/SuicideWatch (49.0) and r/depression (40.8) are less accurate, about 69.4 and 66.0 respectively, but higher true positive rates, 82.1 and 75.0 respectively.

| Metric Category    | Metric      | M1 Std | M2 Std | M3 Std | Most Stable |
|--------------------|-------------|--------|--------|--------|-------------|
| <b>Performance</b> | Accuracy    | 0.054  | 0.012  | 0.024  | M2          |
|                    | F1-Macro    | 0.021  | 0.008  | 0.015  | M2          |
|                    | F1-Positive | 0.029  | 0.018  | 0.021  | M2          |
|                    | AUC-ROC     | 0.014  | 0.006  | 0.005  | M3          |
| <b>Fairness</b>    | Mean DPD    | 0.045  | 0.033  | 0.038  | M2          |
|                    | Mean EOG    | 0.042  | 0.027  | 0.050  | M2          |
|                    | Mean APG    | 0.030  | 0.024  | 0.015  | M3          |

Table 5.12: Performance by Subreddit Community (M3 Results, Top and Bottom 3) Highest Performing Communities:

| Subreddit            | Sample Count | Accuracy          | TPR   | FPR   | Bias Amp | Crisis % |
|----------------------|--------------|-------------------|-------|-------|----------|----------|
| <b>schizophrenia</b> | 138          | $0.935 \pm 0.008$ | 0.500 | 0.038 | 0.007    | 11.6%    |
| <b>OCD</b>           | 137          | $0.934 \pm 0.012$ | 0.600 | 0.053 | 0.036    | 16.8%    |
| <b>Anxiety</b>       | 166          | $0.934 \pm 0.009$ | 0.400 | 0.032 | -0.006   | 9.6%     |
| <b>depression</b>    | 147          | $0.660 \pm 0.023$ | 0.750 | 0.369 | 0.218    | 40.8%    |
| <b>SuicideWatch</b>  | 157          | $0.694 \pm 0.028$ | 0.821 | 0.347 | 0.217    | 49.0%    |
| <b>BPD</b>           | 146          | $0.753 \pm 0.018$ | 0.636 | 0.226 | 0.137    | 30.8%    |

The tendency of high true positive rates in high-crisis communities can be explained by the fact that an appropriate sensitivity bias exists in high-crisis communities, where the model seems to be more lenient in its ability to predict positive sentiment (hope, recovery, coping) in communities where such posts are anticipated to be more common. The false positive rates are also relatively high in these communities (0.347 r/SuicideWatch, 0.369 r/depression) to mean that the model also over-predicts the positive sentiment in such high-crisis situations. This trade-off is clinically appropriate or not depending on the specific case of application and the relative cost of false positives and false negatives in crisis-oriented communities.

## CHAPTER 6

### CONCLUSION AND RECOMMENDATIONS

#### 6.1 Introduction

In this chapter, the findings of the experiments are brought together as well as the meaning of the findings. The three research goals that were presented in Chapter 1 are the organising frame: compares the performance of architectures, compares fairness of architectures through demographic properties, and compares relationship between the two. This discussion uses the results of Chapter 5 and compares them to the literature results of Chapter 2. Constrained is discussed in an honest manner and the chapter is concluded with recommendations to researchers and practitioners who may follow this work.

All of it is motivated by the need to mention it briefly. The mental health system in Nepal is available to only a limited proportion of those requiring it: 2020 National Mental Health Survey revealed that only 22.7% of adults needing mental help sought any assistance, and 3.7% of people with a mental condition obtained hospital-level care [3]. Sentiment analysis of social media text through automated methods has been mentioned as one of the means to address that gap to some extent. The issue is that BERT-based models have demographic biases in pre-training that lead to unequal classification results, and in the environment where there is practically no way to check model errors by humans, these biases do not serve to fill a real gap, they are actually harmful. The thesis addressed that issue in two ways, by building RMH-Bias-10K, the first demographically annotated mental health sentiment dataset ever, and conducting a controlled study of FairBERT (M3) in comparison with two baselines, to answer the question of whether fairness-aware training is effective at all.

#### 6.2 Summary of Findings

Under the first objective, which is predictive performance, M3 performed better than both baselines in all the main measures. Mean accuracy was 84.0 and macro-F1 0.688. The difference between M1 and M1 is quite high: 10.3 in terms of accuracy, 11.7 in terms of macro-F1, significant on all major metrics with the effect sizes of  $d_z = 1.48$  to  $d_z = 8.09$ . The M2 difference is not so significant yet still significant: 4.5 on accuracy ( $p = 0.024$ ,  $d = 1.59$ ) and 2.8 on macro-F1 ( $p = 0.029$ ,  $d = 1.48$ ). The differences between AUC-ROC were

not significant, which is not surprising, M2 was already a powerful discriminator, and the difference was already small.

The story of the minority class is more subtle. M3 has a positive-class F1 of 0.471, which is an improvement of 53.4 (53.4 percent) and 7.8 (7.8 percent) relative to M1 and M2 respectively. The most practically applicable result of the performance side is getting such a minority-class improvement under a 7.4:1 class imbalance, as well as with majority-class performance. It is a tradeoff that M2 has greater positive-class recall (0.675 vs. 0.618 on M3) but at the expense of about 35 more false positives per evaluation run. Neither of these operating points is always superior, it depends on which is more expensive in a specific deployment environment, either case missing or false alert being generated.

On the second goal of fairness, the outcomes are not clean but are honest. In terms of dialect variation, all three models were already at their lowest on DPD (at 0.053 and less), and M3 brought it down to 0.037. However, dialect-based equalized odds and accuracy parity gap deteriorated slightly when moving to M3 as compared to M2, a direct manifestation of the impossibility result of Barocas et al. [37]: when one criterion of fairness is pushed down, another criterion of fairness can be pushed up. In subreddit affiliation, M3 generated moderate changes -15.6% decrease in equalized odds gap and 14.7% in accuracy parity gap. There was still a significant variance at the community level, as should be anticipated due to the disparity between the language and prevalence of crisis between subreddits.

The more problematic cases were gender and severity of crisis. The gender DPD of M3 0.203 was not less than that of M1 0.196. Posts with the female inference continued to have low accuracy (73.2%), and high rates of false positives than those with a male inference (82.9%). This should be put into context: the number of female-inferred posts in the test set is about 56 not a valid sample to estimate group-specific error rates and the training set is also unbalanced, which constrains what the adversarial component has to work with. The failure of fairness in this case is more of a data issue than a technique issue. The data could be made more gender balanced to provide more to work with the same methods.

An outcome that deserves special notification: on a number of attributes, M2 actually performed better on mean DPD and accuracy parity gap than M1. A more powerful model exacerbated rather than improved the issue of demographic disparities over a more basic

baseline that was trained solely on in-domain data. This is important in practical terms since it challenges the premise that improved performance necessarily leads to equal results. Training BERT without the use of any intervention that encourages fairness on domain-specific data will serve to increase the biases that BERT learnt during pre-training instead of diminishing them.

On the third objective, which is the performance-fairness trade-off, M3 scored the highest on all three primary performance metrics and all three-fairness metrics all at the same time. It is Pareto dominance: no point is better than M3 in any of the dimensions. This is in contrast to the conventional wisdom that accuracy and fairness are the opposing forces. A possible reason is that adversarial debiasing encouraged the encoder to abandon demographic shortcuts that do not generalize well and the model paid more attention to features that are relevant to sentiment. Counterfactual augmentation minimized demographic trend overfitting. Focus loss forced training to concentrate on hard ambiguous cases which are the very cases where demographic bias is likely to appear. It appears that these three interventions have enhanced their generalization and fairness in combination as opposed to exchanging each other.

With that said, this outcome should not be over-generalized. The lack of a trade-off is a fact that is related to this particular dataset and setting. The results of the gender and severity of crisis attributes demonstrates that data scarcity may supersede the advantage of the technique combination. It would be beyond the capacity of this study to argue that fairness intervention is always beneficial in performance.

### **6.3 Limitations**

The most severe limitation is the one of the gender evaluations. A small number of around 56 female-inferred posts in the test set, or approximately 3.7 percent of the test data, means that the estimates of the accuracy and error rate in the group are statistically unreliable. This is complicated by the 65 e-percent unknown-gender rate in the data: most posts had no discernible gender signal whatsoever, and thus the female-inferred category is a tiny and possibly unrepresentative sample of a larger unknown population. The results of this study on gender fairness must be viewed as trial results that will encourage the data collection process and not as the proven conclusions.

The four demographic attribute labels are all textually based instead of being self-reported. The gender assumption is based on a binary system, which eliminates non-binary and transgender identity. Dialect is narrowed down to a standard vs. non-standard dichotomy which simplifies the real sociolinguistic complexity. VADER and key word matching are used to estimate crisis severity as compared to clinical assessment. These are a accepted fairness research practice in the absence of self-reported demographics [32], but the metrics obtained are approximations. Labelling noise indicates that the measures of fairness are, in some instances, understating the actual disparities and are, in other cases, overstating the disparities.

The data sample is English monolingual and based on the Reddit platform, the user base is biased with young, western and technologically oriented users. The transferability of these findings to other platforms, or even other languages, or even mental health communities which use Nepali as their language, the context in which the research was initially motivated, is beyond the scope of this study. The Reddit audience is not reflective of the individuals that are most vulnerable to the treatment gaps reported in the National Mental Health Survey of 2020.

The four demographic attributes are considered to be independent during analysis. Individual persons hold overlapping roles throughout all of them simultaneously, and even individual effects of compounding may not always resemble impacts of individual attributes. At the existing dataset size, intersectional analysis has not been possible since the intersection cells would have a very small number of instances, and any evaluation of them would not be very reliable.

Lastly, it has no clinical validation. All the results presented in this thesis are obtained as the result of offline assessment of experimental conditions. None of the predictions was reviewed by practitioners. There were no real users who were used. The existence of the technical fairness gains in a real deployment to more equitable service delivery is an entirely open question that this study cannot answer.

## **6.4 Future Research Directions**

### **6.4.1 Dataset Expansion and Demographic Representation**

The simplest method to deal with a number of the present limitations is to create a larger and more demographically balanced annotated corpus. Indifferently scraping Reddit reproduces the current demographic bias of the site. A more proactive recruitment effort among underrepresented groups, or sourcing via systems with dissimilar user’s crisis text lines, telehealth systems, community mental health programs, would provide better training and evaluation value on groups that have insufficient cases at present. In the case of Nepal, the fairness-conscious methodologies used in this paper would apply to the population that the study is intended to serve by creating annotated mental health datasets in Nepali and the local languages of the region. Fairness evaluation would be much more credible, however, when self-reported demographics are incorporated where possible instead of being inferred. It would mean improvements to methodology per se to work with platform providers where demographics are collected with informed consent, or to design studies with voluntary disclosure of demographics.

#### **6.4.2 Intersectional Fairness Analysis**

The next area of work should look into the way in which combinations of safeguarded features strictly affect model behavior. When someone who speaks non-standard English writes a post in a high-crisis sub-subreddit, there is the danger of compounded prediction difference that neither account explains independently. This type of intersectional analysis is very demanding in terms of sample sizes per intersection cell compared to the current data. Technically, a logical follow-up of adversarial debiasing, which seeks to control the distribution of multiple attributes simultaneously (instead of a single attribute at a time) is also available to the literature on fairness, but has not been applied to mental health NLP yet.

#### **6.4.3 Architectural Extensions and Alternative Techniques**

There was one model family that was subjected to one set of fairness techniques in this thesis. A further extension to the domain-adapted pre-trained models, bigger variants of transformers, and GPT-family models would help understand whether the dynamics here are applicable to model families. The association between the scale of a model and the demographic bias in the specialized areas is not well-known. The

other mitigation strategies, which include reweighting, post-hoc calibration, and prompt-based debiasing of large language models, should all be experimented in this regard. The issue of the most effective combinations of techniques that yield maximum positive joint performance-fairness results in NLP of the mental profession is not yet closed.

#### **6.4.4 Prospective Clinical Validation**

Clinical validation in a realistic deployment environment is the most critical step that follows. It implies collaboration with practitioners, evaluating whether the model outputs can be interpreted and taken into practice, and assessing whether the groups that were found underserved by the technical evaluation get better supported in the case when the fairness-aware model is used. In the absence of such validation, the findings here are not evidence of a benefit to public health, but are technically the findings. The same is also the most difficult step, and it involves institutional collaboration, the ethical approval, and a wise study design, however, it is the sole way a research input can be transformed into a system that can be responsibly applied to real people.

### **6.5 Recommendations**

#### **6.5.1 Recommendations for Researchers**

Fairness assessment must be an inherent aspect of mental health NLP studies rather than one imposed upon them by those who request them to do so. Research that gives aggregate performance is not able to inform whether its model is equally effective among demographic groups. Since the lack of equality in mental health applications has direct implications on vulnerable persons, it is difficult to endorse that omission. The assessment system applied within this thesis three fairness measures, four guarded attributes, multi-seed replication and statistical significance testing provides a basic template that can be changed to different languages, platforms and demographic aspects. Annotated datasets that contain demographic metadata are to be considered a field-level priority. The primary constraint of fairness work in mental health NLP currently is the lack of such datasets. One of the approaches that can be

developed or modified by other approaches is the two-stage annotation methodology of Chapter 3, which is documented with independent expert validation and formal reliability measurement. By making such datasets open as opposed to holding them internally, one would increase developments in the field by a significant margin.

### **6.5.2 Recommendations for Practitioners and System Designers**

The organizations need to insist on evidence of fair performance across the demographic groups in their target population before they deploy any automated sentiment analysis tool to monitor their mental health condition not after a pilot, but as a condition of deployment. A high score on an independent test set held out is not a sufficient condition to follow that a system will be fair to respond to a diverse population. The acceptance criteria must include subgroup level assessment of the attributes which are most applicable to the context.

Human supervision is not an option. The error rates that M3 attains, although being an evident improvement upon the baselines, are too large to use the system fully automatically when it comes to making decisions concerning mental health status. These tools are supposed to act as a triage tool that presents cases to practitioners and not as a resource distribution tool. This is particularly true of mental health, as a false positive (false alarms about a person in distress) and false negatives (person requiring assistance missed) are both costly in literal terms to the lives of real people.

Justice is not something that remains unchanged after deployment. The demographics of users can vary, the languages can change and the dynamics of a platform can vary in a manner that may change the manifestation of demographic differences at work. Measures of evaluation offline will not be a complete forecast of what occurs operationally. Monitoring systems to measure the performance of the subgroups with time and feedback systems to allow the practitioners to identify cases that appear biased should be part of any deployed system. It is much easier to integrate that infrastructure into the system during its initial construction rather than doing it later when the issues arise.

## **6.6 Concluding Remarks**

This research had a concrete problem as its starting point rather than an abstract problem. The access to mental health care in Nepal is a limited percentage of the population in need of services, and automated text analysis tools have been suggested as one of the solutions. The issue that inspired this thesis was the fact that those tools are not neutral, they have biases that may only reinstate the inequities they are intended to remedy and especially in the environment where one has no human skill to scrutinize and rectify the mistakes.

The two contributions here are a response to that problem in different levels. RMH-Bias-10K contains the dataset that contains the demographic labeling required to estimate the fairness issue of the mental health sentiment analysis, a task that was previously impossible to do using the existing datasets. This three-model comparison demonstrates that adversarial debiasing, counterfactual data augmentation, and focal loss can be used to decrease demographic differences and preserve or increase predictive accuracy, given the conditions examined in this paper.

The conclusion that M3 is better and fairer at the same time is really positive. Nonetheless, the shortcomings are not imaginary and the honesty of reporting them is as important as the honesty of reporting the outcomes. The data is limited which limits the gender assessment. The demographic labels are not ground truth but they are inferred proxies. Only the English Reddit is included in the dataset. It has no clinical validation. These are serious caveats these are what the findings are able and unable to support. The findings create a positive path of direction, not an ultimate solution.

The foundation this thesis lays down is as follows: a demographically annotated corpus, a controlled multi-model comparison, a multi-metric fairness evaluation framework and a tested combination of fairness-conscious training methods used on mental health NLP in the first instance. The basis of that is constructed in order to be developed. This would all require larger datasets, intersectional analysis, wider architectural comparisons and finally clinical validation before any of this could be used to support actual deployment. Hopefully, work here can hasten that follow-up research, and in some small respect, lead toward the goal of ensuring that AI-assisted mental health tools are fair to all, and where the probability of error is the lowest and the need is the greatest, the resource-constrained environment.

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## APPENDIX A

### COLAB NOTEBOOK LINKS

The following Google Colab notebooks contain all code necessary to load, inspect, preprocess, and prepare the RMH-Bias-10K dataset for the experiments described in this thesis. Each notebook is self-contained, includes markdown explanations, and can be run end-to-end in a free Google Colab instance.

All notebooks use fixed seeds for reproducibility (seed= [42, 123, 456, 789, 1024]). You can change the seed or paths to experiment further. The dataset CSVs are assumed to be placed in the Colab/content/folder or mounted from Google Drive.

#### A.1 RMH-Bias-10K – Dataset Loading & Basic Statistics

Link:[https://colab.research.google.com/drive/1Q9UrHcY2pHmpZcVFgj0BECIJWBimRF XO?usp=share\\_link](https://colab.research.google.com/drive/1Q9UrHcY2pHmpZcVFgj0BECIJWBimRF XO?usp=share_link).

Link:[https://colab.research.google.com/drive/1cXBiaMF0zaPd5vHsZ7IGZZAO6TWCgUqz?usp=share\\_link](https://colab.research.google.com/drive/1cXBiaMF0zaPd5vHsZ7IGZZAO6TWCgUqz?usp=share_link).

Link to the dataset RMH-Bias-10K.csv and RMH-Bias-10K-Annotated.xlsx

[https://drive.google.com/drive/folders/1UNXJWNEjpJ2LcU8VIIYalsPY9ZuyEOEV?usp=share\\_link](https://drive.google.com/drive/folders/1UNXJWNEjpJ2LcU8VIIYalsPY9ZuyEOEV?usp=share_link).

#### A.2 RMH-Bias-10K – Preprocessing & Train/Val/Test Split

Link:[https://colab.research.google.com/drive/1A9gXsXUIn2Egq6APg59\\_E0R3DTxMWn6t](https://colab.research.google.com/drive/1A9gXsXUIn2Egq6APg59_E0R3DTxMWn6t).

#### A.3 Model Training – BiLSTM (M1), BERT and FairBERT (M2 & M3)

Link:

<https://colab.research.google.com/drive/1cXBiaMF0zaPd5vHsZ7IGZZAO6TWCgUqz#scrollTo=eOV1vpEMzjV6>.

## APPENDIX B

### FAIRNESS METRICS FORMALIZATION

This appendix formalizes the fairness metrics used in the thesis and provides worked examples to clarify interpretation.

#### B.1 Notation

Let:

$Y \in \{0,1\}$  denote the ground-truth label (0 = negative class)

$\hat{Y} \in \{0,1\}$  denote the model prediction (0 = negative class)

$A \in \{\text{female}, \text{male}\}$  denote the (inferred) gender group

#### B.2 Demographic Parity Difference

Demographic Parity Difference (DPD) for the negative class is defined as:

$$\text{DPD} = P(\hat{Y} = 1 \mid A = \text{female}) - P(\hat{Y} = 1 \mid A = \text{male})$$

A positive value indicates that the model predicts negative sentiment more often for the female group, conditional on no other information. Perfect demographic parity requires  $\text{DPD} = 0$ .

#### B.3 Equalized Odds

Let:

$$\text{TPR}_{\text{g}} = P(\hat{Y} = 1 \mid Y = 1, A = g) \text{ (True Positive Rate for group } g)$$

$$\text{FPR}_{\text{g}} = P(\hat{Y} = 1 \mid Y = 0, A = g) \text{ (False Positive Rate for group } g)$$

for group  $g \in \{\text{female}, \text{male}\}$ .

The Equalized Odds gaps are:

$$\text{EO-TPR} = \text{TPR}_{\text{female}} - \text{TPR}_{\text{male}}$$

$$\text{EO-FPR} = \text{FPR}_{\text{female}} - \text{FPR}_{\text{male}}$$

Values close to zero indicate more similar error behavior between groups. Perfect equalized odds requires both gaps equal zero.

#### B.4 Accuracy Parity Gap

Accuracy Parity Gap (APG) measures how much the predictive distribution diverges from the true label distribution across groups:

$$BA = [P(\hat{Y} = 1 \mid A = \text{female}) - P(Y = 1 \mid A = \text{female})] - [P(\hat{Y} = 1 \mid A = \text{male}) - P(Y = 1 \mid A = \text{male})]$$

A positive BA indicates that existing disparities in negative labels are being amplified by the model. BA = 0 indicates the model preserves the baseline disparity without amplification.

### B.5 Illustrative Example

Suppose that on the test set:

$$P(Y = 1 \mid A = \text{female}) = 0.70$$

$$P(Y = 1 \mid A = \text{male}) = 0.68$$

$$P(\hat{Y} = 1 \mid A = \text{female}) = 0.78$$

$$P(\hat{Y} = 1 \mid A = \text{male}) = 0.70$$

Then:

$$DPD = 0.78 - 0.70 = 0.08$$

$$BA = (0.78 - 0.70) - (0.70 - 0.68) = 0.08 - 0.02 = 0.06$$

These values suggest a modest amplification of gender disparity in negative predictions relative to the underlying label distribution. The model predicts negative sentiment 8 percentage points more often for females, and this represents a 6 percentage point amplification beyond the baseline difference in true labels.

## **APPENDIX C**

### **ETHICAL CONSIDERATIONS AND DATA HANDLING**

This appendix supplements the ethical discussion in the main chapters by summarizing key steps taken to mitigate risks associated with working on mental-health-related user-generated content.

#### **C.1 Anonymization and Paraphrasing**

All example posts shown in this thesis are anonymized and paraphrased. Direct usernames, links, and identifiable details are removed or obfuscated. The aim is to preserve linguistic characteristics relevant to modeling while protecting the privacy of the original authors.

Where example posts are presented, they represent composite or synthetic examples that reflect typical patterns in the data rather than verbatim quotes from individual users.

#### **C.2 Compliance with Platform Policies**

Data was collected only from publicly accessible Reddit communities and used strictly for non-commercial academic research. Care was taken to respect Reddit's terms of service and community guidelines, including:

- Not attempting to deanonymize or contact individual users
- Restricting data collection to public posts
- Using the data exclusively for research purposes
- Not redistributing raw data without appropriate permissions

#### **C.3 Informed Consent and Public Data**

While Reddit posts are publicly available, users may not anticipate their content being used in research. This study follows the principle that:

- Public online content can be used for research with appropriate anonymization
- Individual posts are never attributed to specific users
- Sensitive content is handled with particular care
- The aggregated nature of the analysis minimizes individual privacy risks